CSCI 4623/5623 **DIGITAL FORENSICS**

THE UNIVERSITY of NEW ORLEANS

# 01 CONCEPTS Review: OS & NETWORKING

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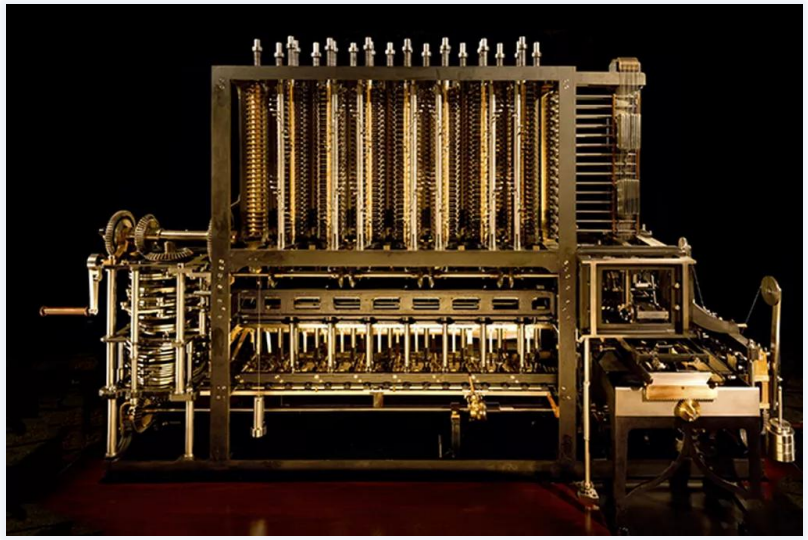
# What is a computer?

A brief history of computing

# The Early Days

Antikythera mechanism
(150-100 BC)



Difference Engine No.2
(design: 1847-49)

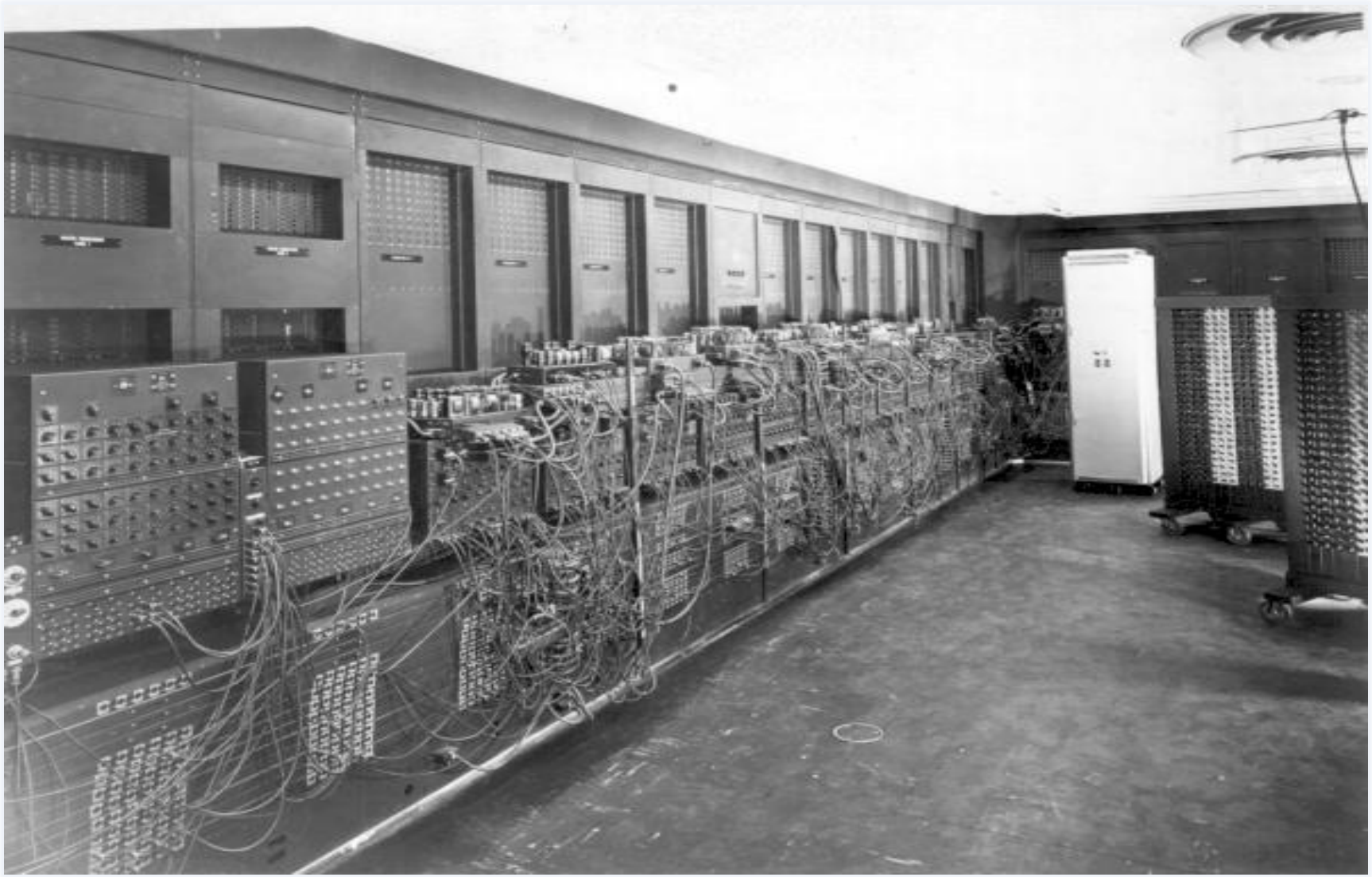
Burroughs desktop calculator
ca.1910

Ada Lovelace (nee Byron)
First Programmer

# Human Computers

Dryden Flight Research Center E49-0053 Photographed 10/49
Early "computers" at work. NASA photo

# ENIAC (MID-1940's)



# ENIAC: THE DETAILS

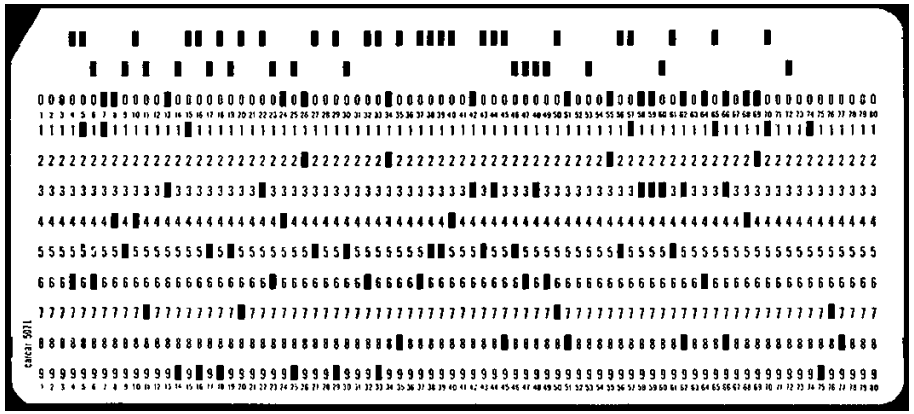
- Built to compute bomb trajectories and replace ***human computers***
- Almost 20,000 vacuum tubes
- Thousands of  square feet to house
- 150,000 watts of power
- 30 tons!!
- Specs
  - 5,000 additions per second
  - Could do complicated calculations requiring 2 days for a human “computer” in 30 seconds
- Your Chromebook: ***billions*** of operations per second!

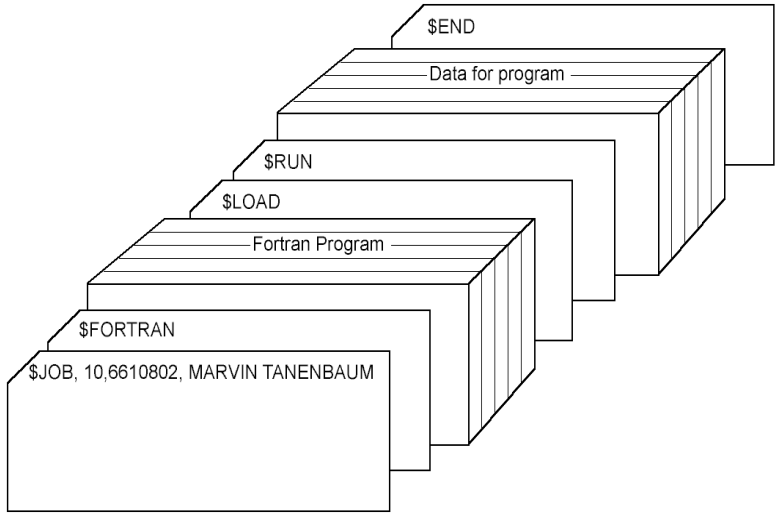
# UNIVAC (EARLY 1950S)

# UNIVAC: THE DETAILS

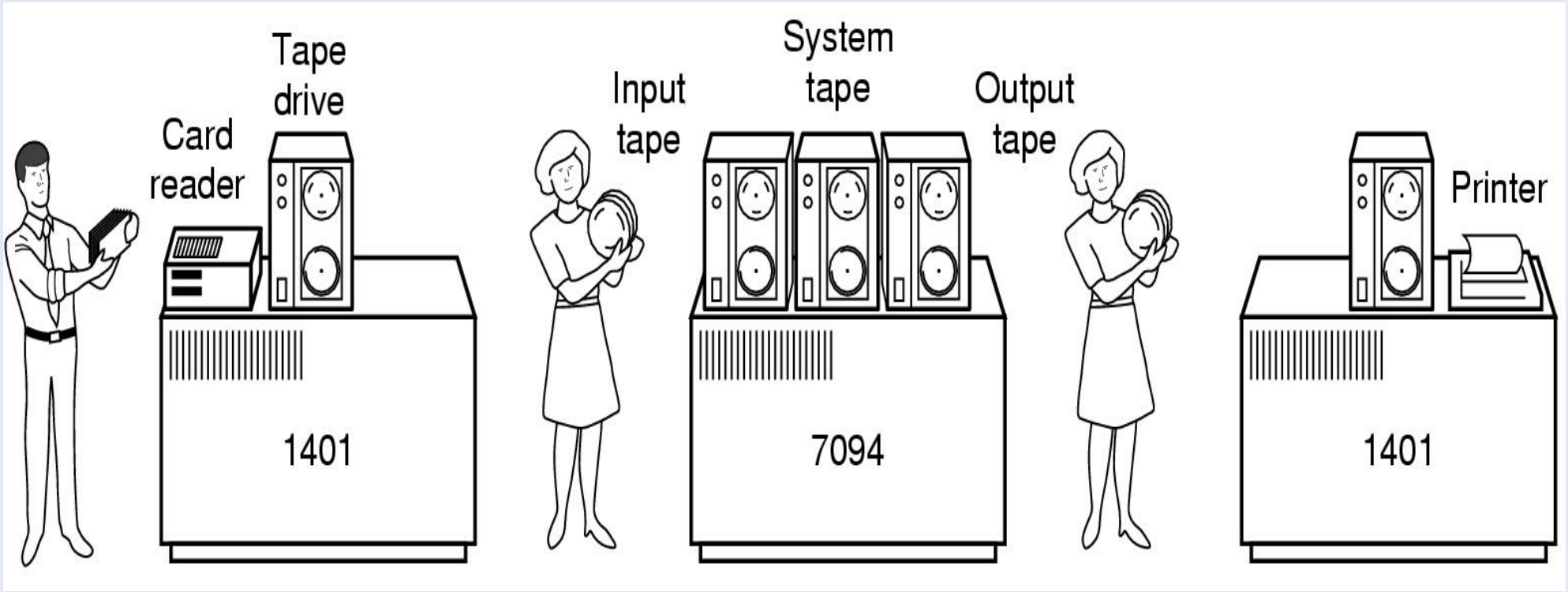
- Walk-in design
- 5000+ tubes
- 8 tons
- 300+ square feet
- $1 million
- Over 40 units sold
- Substantially more reliable than ENIAC
- Stored program computer—no more pulling wires!
- Predicted Eisenhower's landslide electoral victory over Adlai E. Stevenson in 1952!

# I/O Improvement: punch cards



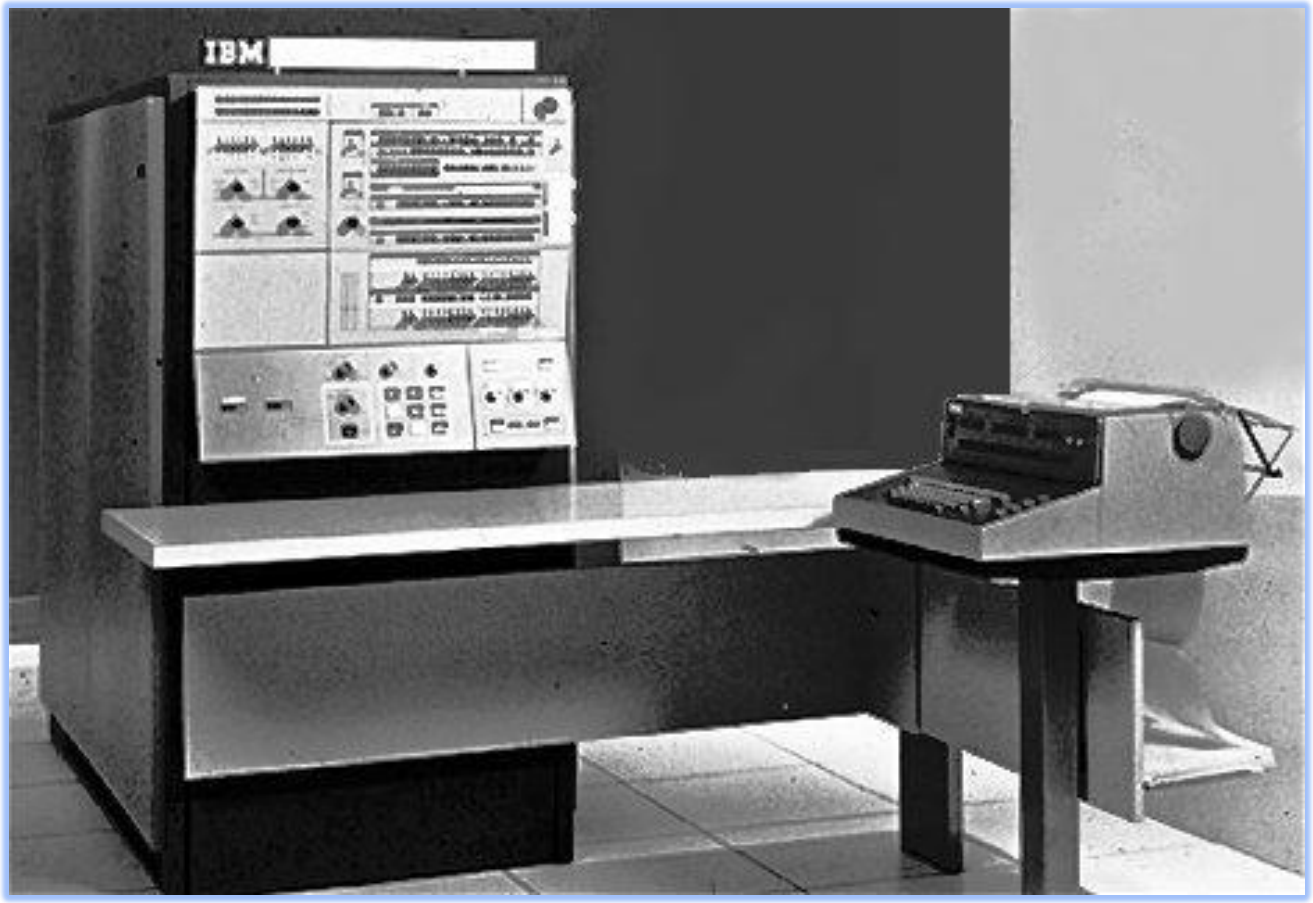


# Batch Systems




Human labor used to optimize CPU utilization

# IBM 360 series (mid-1960s)




First modern operating system:

**OS/360**

# IBM 360: THE DETAILS

- Price: hundreds of thousands to $1million+
- Basis of computers sold through the 1990s
  - 370 series (1970, still compatible with 360)
  - 390 series (still in use)
- Wildly popular—gave IBM domination over the computer market
- Sold 14,000 units by 1968
- By 1975, IBM and “everyone else”
- Tens of millions of operations per second
- Very stable
- Modern units, less than 5 minutes down per year

# MULTICS: AN OS REVOLUTION

- MULTICS (http://www.multicians.org)
  - **MULT**iplexed **I**nformation and **C**omputing **S**ervice
- Goals
  - Convenient remote terminal use.
  - Continuous operation analogous to a utility
  - A wide range of system configurations, changeable without system or user program reorganization.
  - A high reliability internal file system.
  - Support for selective information sharing.
  - Hierarchical structures of information for system administration
  - Support for a wide range of applications.
  - Support for multiple programming environments & human interfaces.
  - The ability to evolve the system with changes in technology

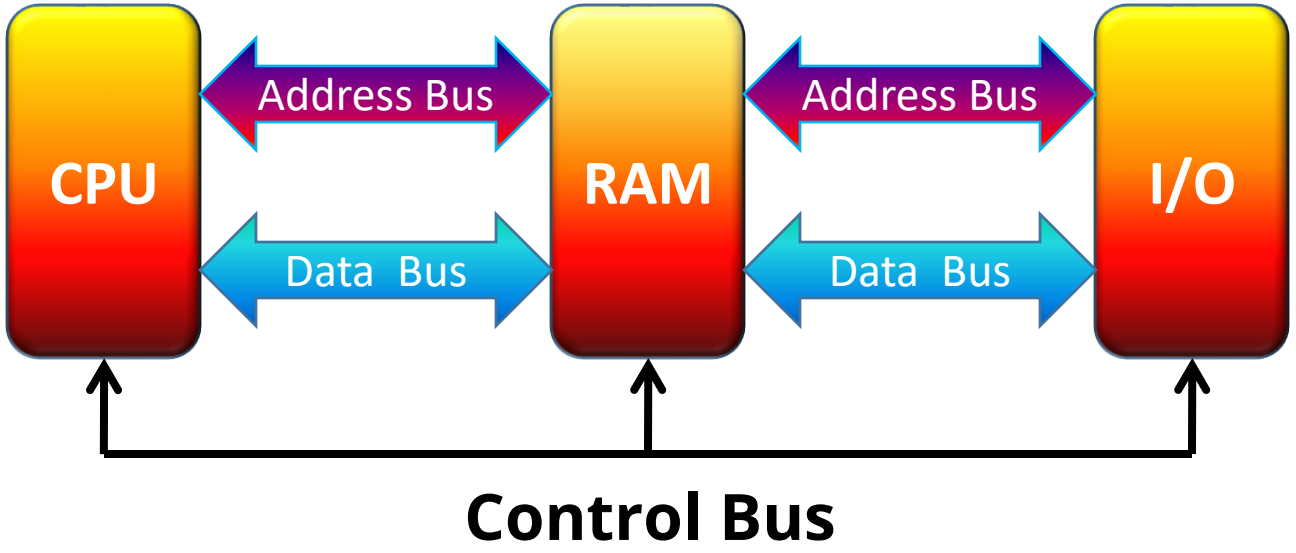
# Further OS Evolution

- Minicomputers (PDP – [1-11])
- **UNIX**
  - Ken Thompson (Bell Labs)
  - System V (AT & T)
  - BSD (Berkeley)
  - POSIX (IEEE standard)
  - MINIX
  - **Linux**
  - …

# OS Evolution

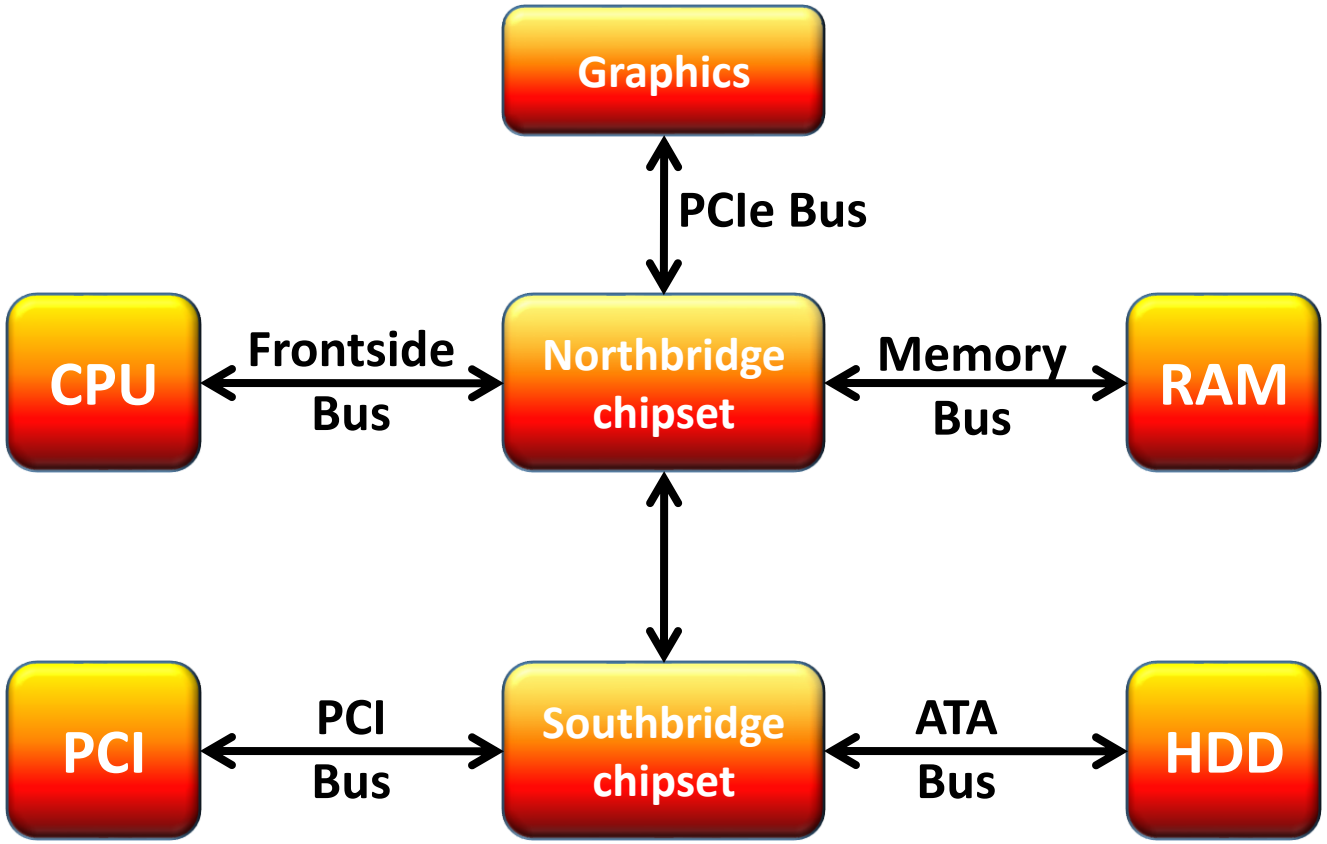
- CP/M (Control Program for Microcomputers)
  - Intel 8080
  - Gary Kildall 1974
- DOS (1981) ➔ MS-DOS
- GUI
  - Doug Engelbart, Xerox PARC, 1960s
  - Mac OS
  - MS Windows XX
  - X Windows

# Conceptual Computer System Architecture




Known as the von Neumann architecture

# Modern *Intel* Architecture (simplified)

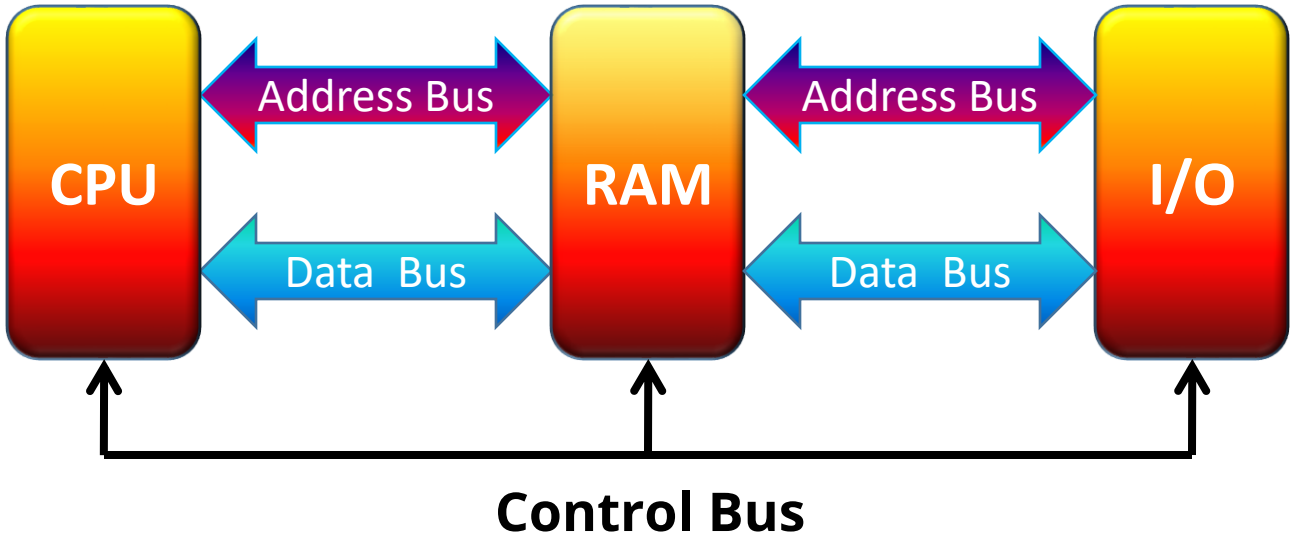


# Main resources the OS needs to manage

***CPU:***
Scheduling of computation

***Main memory (RAM):***
Allocation of memory needed for computation

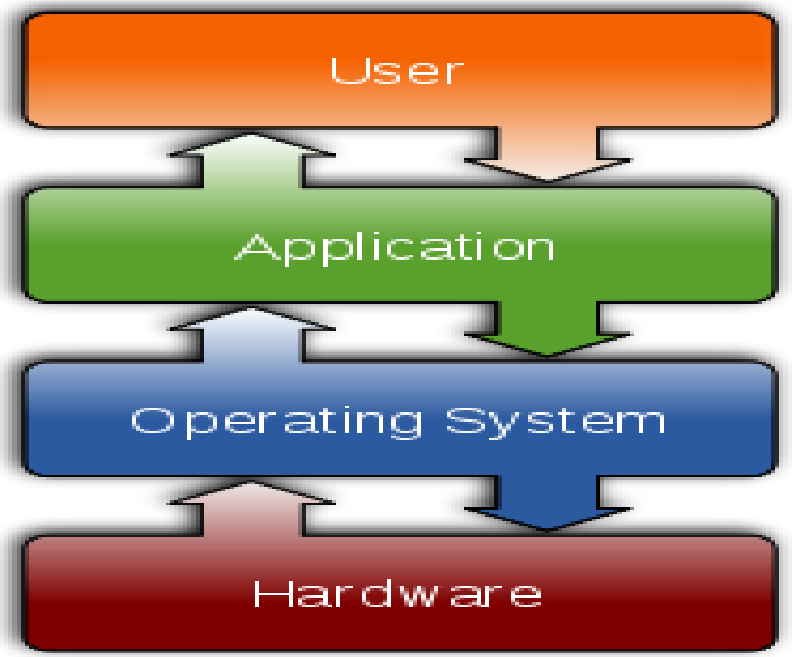
***External devices:***
I/O communication with all devices



# What is an OS?

# Working Definition

*A software intermediary between users/application software and the bare hardware.*



# Goals of OS

- Utilize hardware efficiently and fairly
- Execute user programs and act as a “policeman”
- Make computer system easy (at least *easier*) to use
- Provide some measure of abstraction
  - (Hide the details of the hardware)
- **Provide mechanisms for access control and information protection**

# Hardware Protection

- Automatically enforced on *every* instruction
- Dual-mode operation
  - » Kernel (privileged) mode
  - » User (non-privileged) mode
- Used for
  - » CPU Protection
  - » Memory Protection
  - » I/O Protection

# Dual-Mode Operation

- Sharing system resources requires OS ***policing***.
- How to differentiate between user/OS instructions?
  - Hardware support: (at least) two modes of operation:
    - ***User mode*** – execution done on behalf of a user.
    - ***Monitor/kernel/system mode:*** execution done on behalf of OS
- In kernel mode, all system resources are accessible to the running code
- In user mode, only explicitly allocated resource are available:
  - E.g.: 10ms of CPU time, 17MiB of RAM, …
  - These is limits are strongly enforced

# CPU Protection

- *Timer* – interrupts computer after specified period to ensure operating system maintains control.
  - Timer is decremented every clock tick.
  - When timer reaches the value 0, an interrupt occurs.
- Timer commonly used to implement time sharing.
- Time also used to compute the current time.
- Setting up the timer is a privileged instruction.
  - This ensures that the OS will regain control of the CPU
  - … otherwise, OS/other apps may never execute again (*aka* ***denial of service*** attack, or ***DoS***)

# Process definition

- A ***program*** is
  - a sequence of instructions (a computation) that accomplishes a specific purpose.
  - stored on persistent storage, loaded and executed when necessary.
- A *process/task* is a program in execution.
  - each process gets a (dynamic) allocation of the system's CPU, RAM, and other resources.
  - multiple instances of the same program can co-exist at the same time.
  - by default, each process has its own address space that is protected from those of other processes.
  - the number of processes executing in parallel is limited by available CPUs/cores.
- A process is the basic unit of CPU work

# Processes

Task Manager

File Options View

Processes | Performance | App history | Startup | Users | Details | Services

| Name | 16% CPU | 42% Memory | 3% Disk | 0% Network |
|---|---|---|---|---|
| Firefox (32 bit) | 5.8% | 833.5 MB | 0 MB/s | 0 Mbps |
| System | 2.4% | 0.1 MB | 0.1 MB/s | 0 Mbps |
| Desktop Window Manager | 1.7% | 130.4 MB | 0 MB/s | 0 Mbps |
| GNU Image Manipulation Progr... | 1.6% | 60.4 MB | 0 MB/s | 0 Mbps |
| Snipping Tool | 1.4% | 2.8 MB | 0 MB/s | 0 Mbps |
| Windows Explorer | 1.0% | 86.7 MB | 0.1 MB/s | 0 Mbps |
| Firefox (32 bit) | 0.8% | 7.4 MB | 0 MB/s | 0 Mbps |
| Task Manager | 0.4% | 20.9 MB | 0.1 MB/s | 0 Mbps |
| Dropbox (32 bit) | 0.1% | 125.9 MB | 0 MB/s | 0 Mbps |
| Lenovo.Modern.ImController.Pl... | 0.1% | 21.9 MB | 0 MB/s | 0 Mbps |
| VMware Authorization Service (... | 0.1% | 3.0 MB | 0 MB/s | 0 Mbps |
| System interrupts | 0.1% | 0 MB | 0 MB/s | 0 Mbps |
| Firefox (32 bit) (2) | 0.1% | 203.8 MB | 0 MB/s | 0 Mbps |
| Service Host: Local Service (No ... | 0.1% | 21.0 MB | 0 MB/s | 0 Mbps |

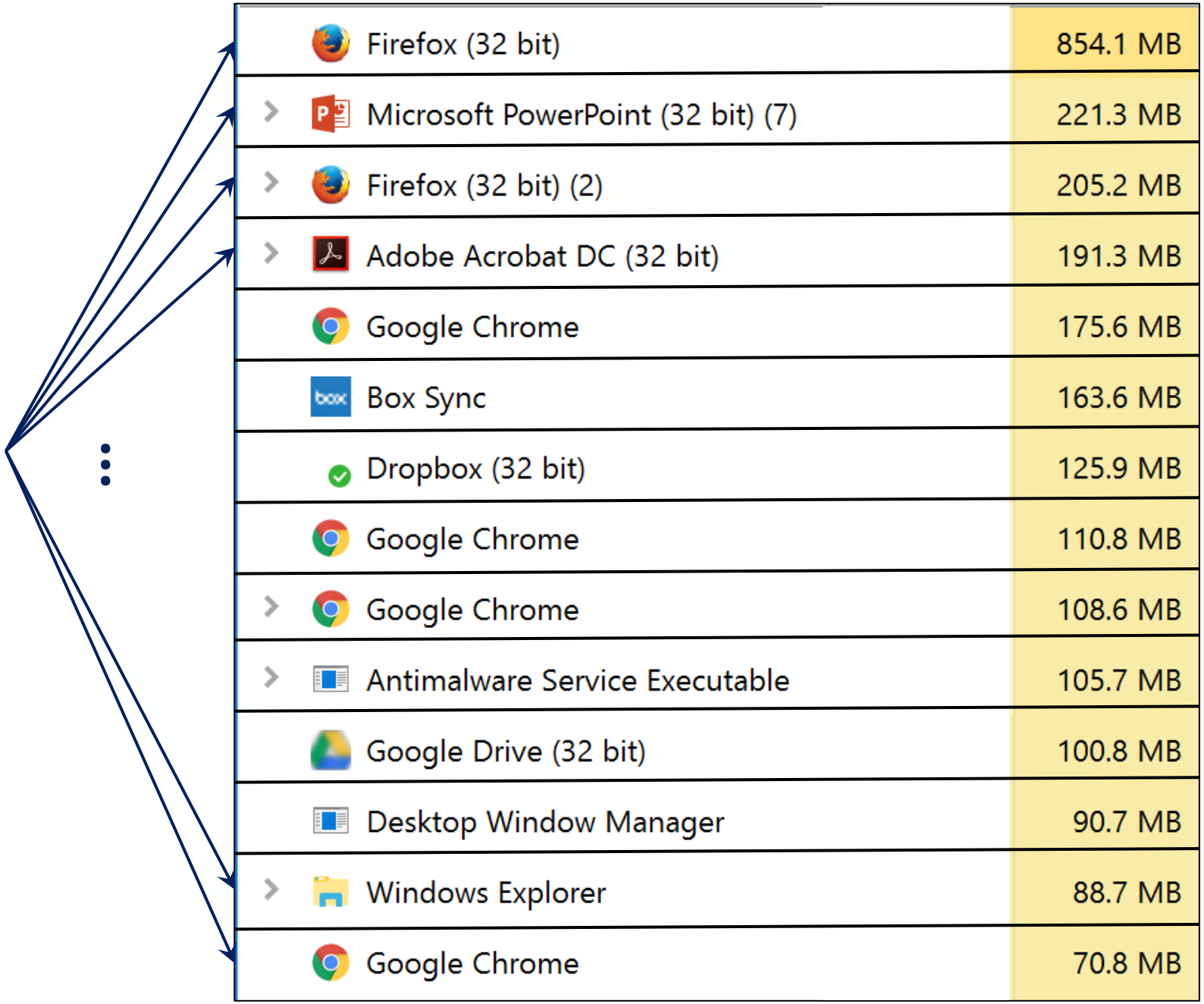
Fewer details

End task

# Main Memory Protection

- Special limit registers
  - Loaded with the allocated address bounds for the process
  - Process is **NOT** allowed to access RAM outside the assigned range
  - Enforced as part of the execution of **EVERY** instruction
- Memory protection provides the basic means of isolating processes from each other. Useful for:
  - Privilege isolation (b/w app processes & OS processes)
  - Fault isolation (process crash)
  - Privacy isolation—a process' data is private by default
- Note: The OS can access all memory

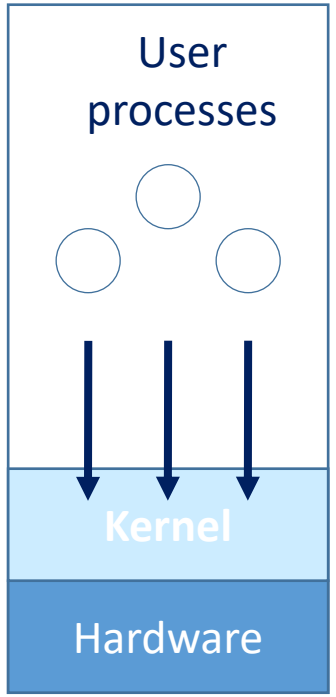
**DISJOINT** RAM allocations
w/ **ENFORCED** boundaries



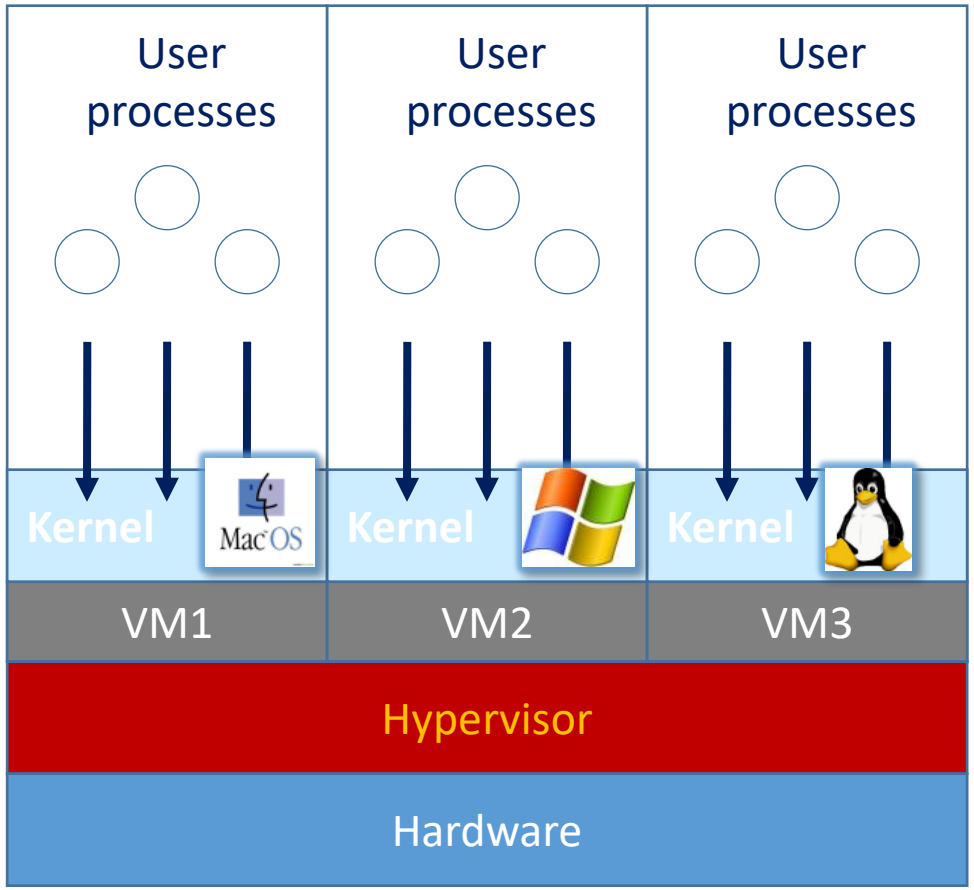
# Virtual Machines

- A ***virtual machine*** provides an interface identical to the underlying bare hardware.
  - The **hypervisor** creates the illusion that a process has its own CPU, RAM, I/O devices.
  - Each guest provided with a (virtual) copy of underlying computer.
  - Hardware support is needed for VMs to run efficiently

# Virtual Machines




Non-virtualized machine:
**Single** OS instance is present




Virtualized machine:
**Multiple** OS instances may be present

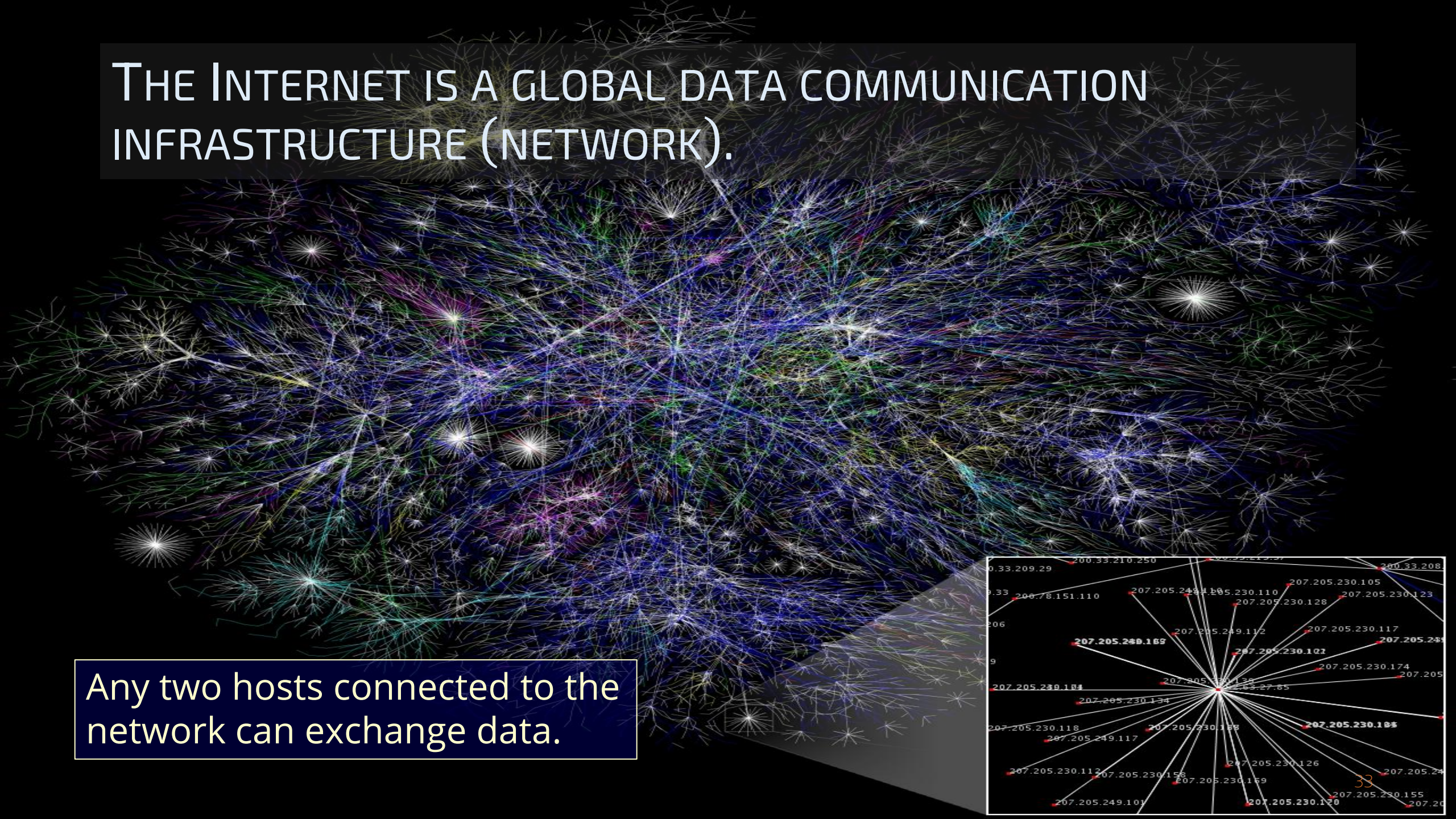
# Concept Index

- Von Neumann computer architecture
- OS definition and computer system layers
- Dual-mode hardware operation
  - Timer-based CPU protection
  - Address-based RAM protection
- Processes as basic units of computational work
- Virtualization & virtual machines

# What is the Internet?

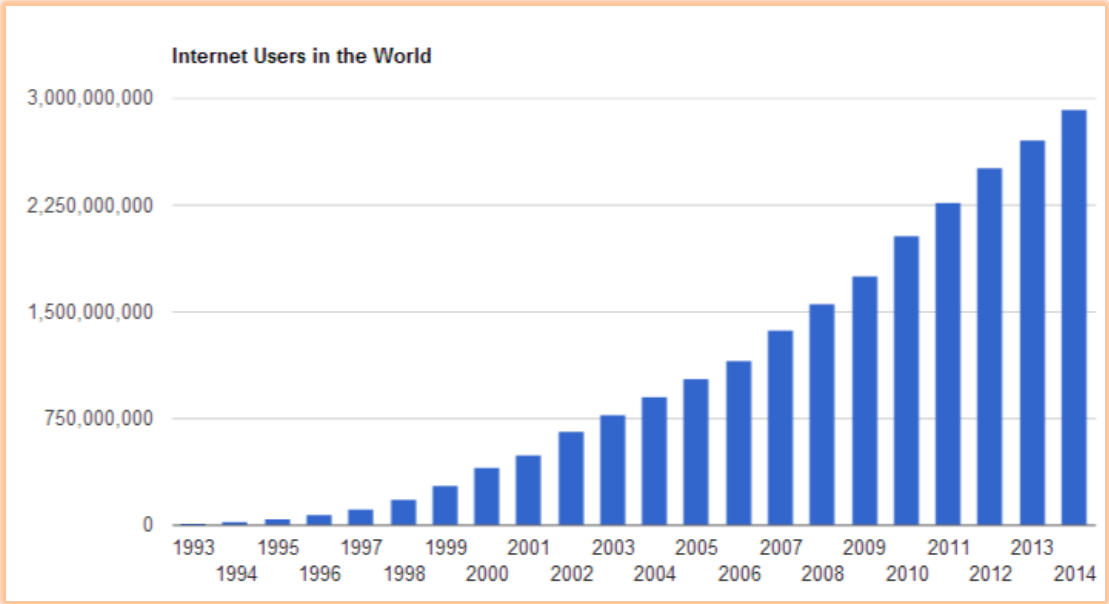
Networking basics

# The Internet is a global data communication infrastructure (network).

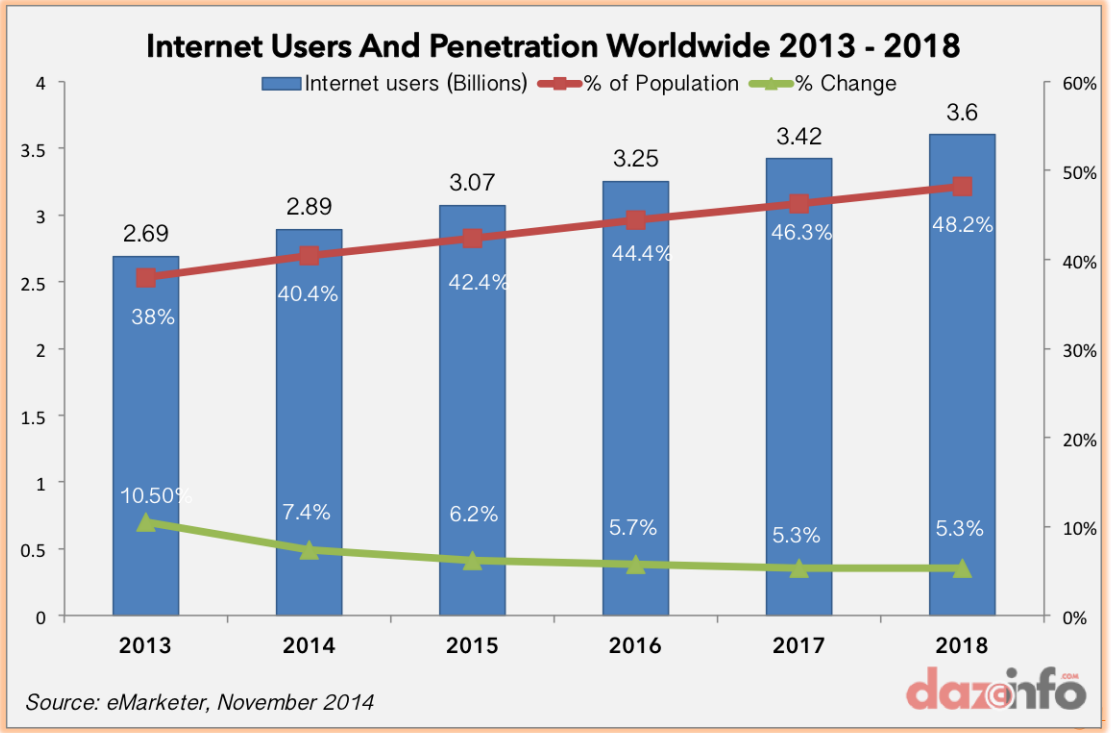



Any two hosts connected to the network can exchange data.

# THE 'NET HAS 3BLN+ USERS

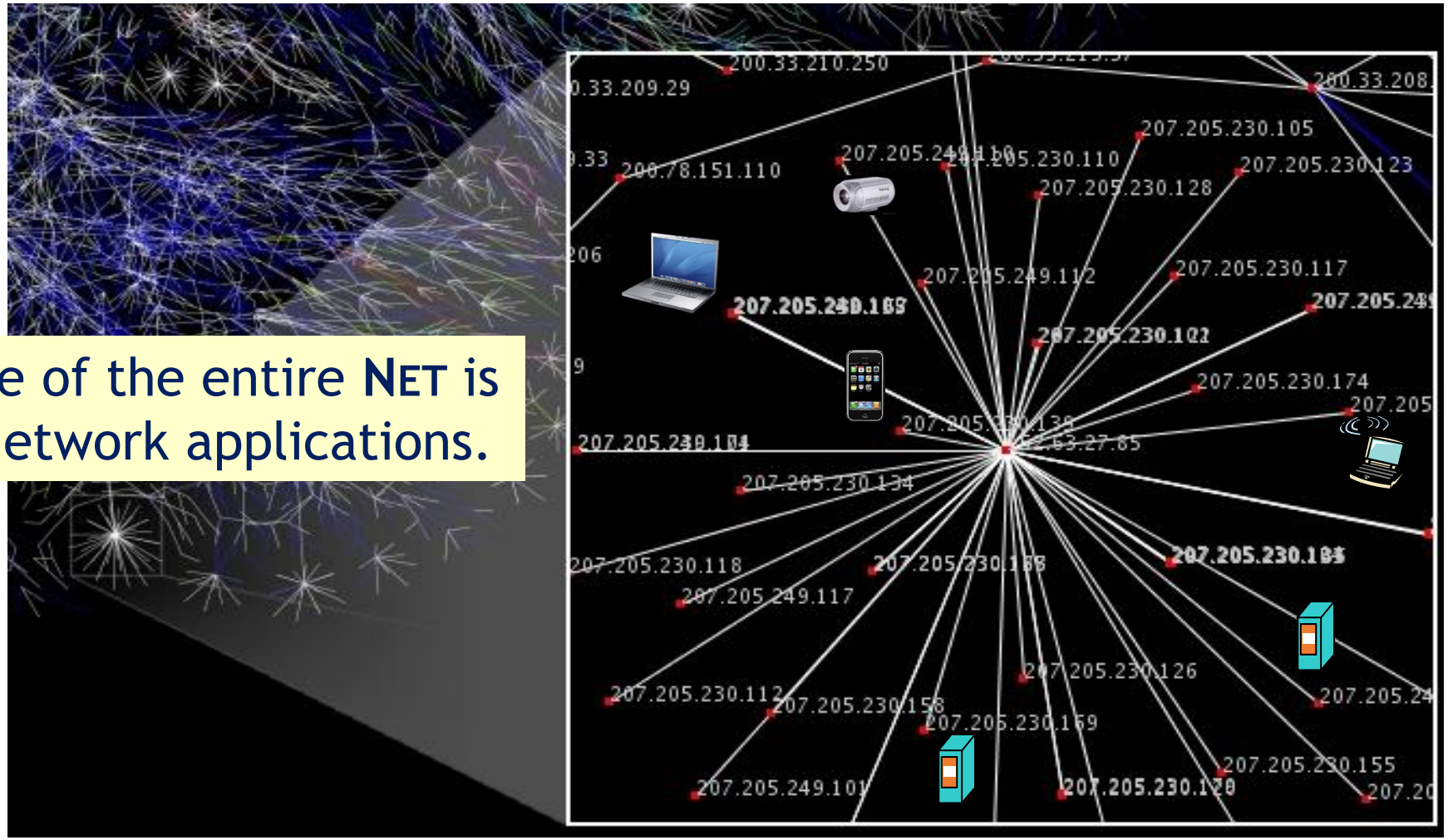



The physical network infrastructure is **EVERYWHERE**; nobody can control all of it.



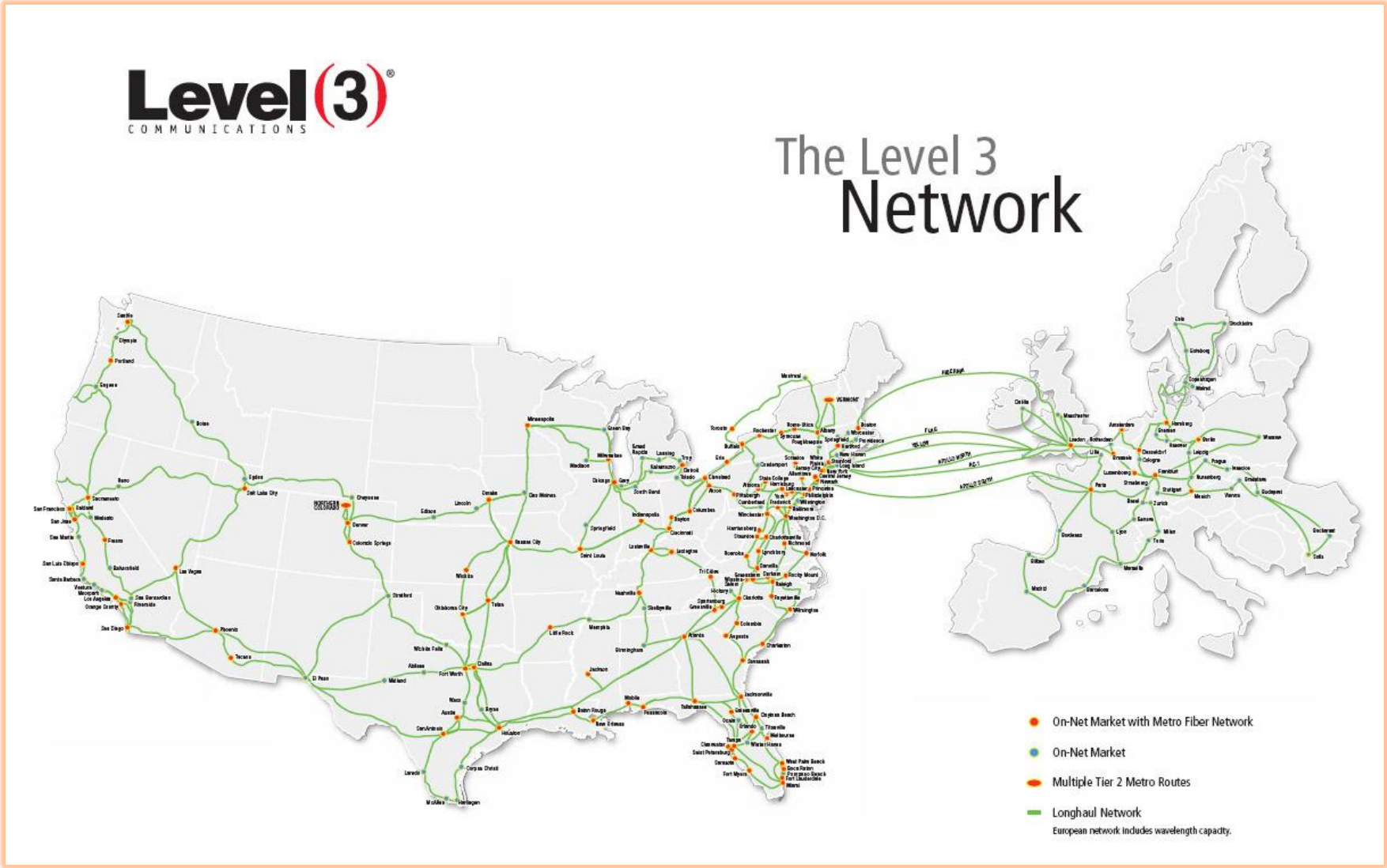

For this work, the world must agree on common standards & protocols; Net management is inherently cooperative.

# THE **NETWORK EDGE** CONSISTS OF ALL THE HOSTS THAT SERVE END USERS VIA NETWORK APPLICATIONS.




The purpose of the entire **NET** is to enable network applications.

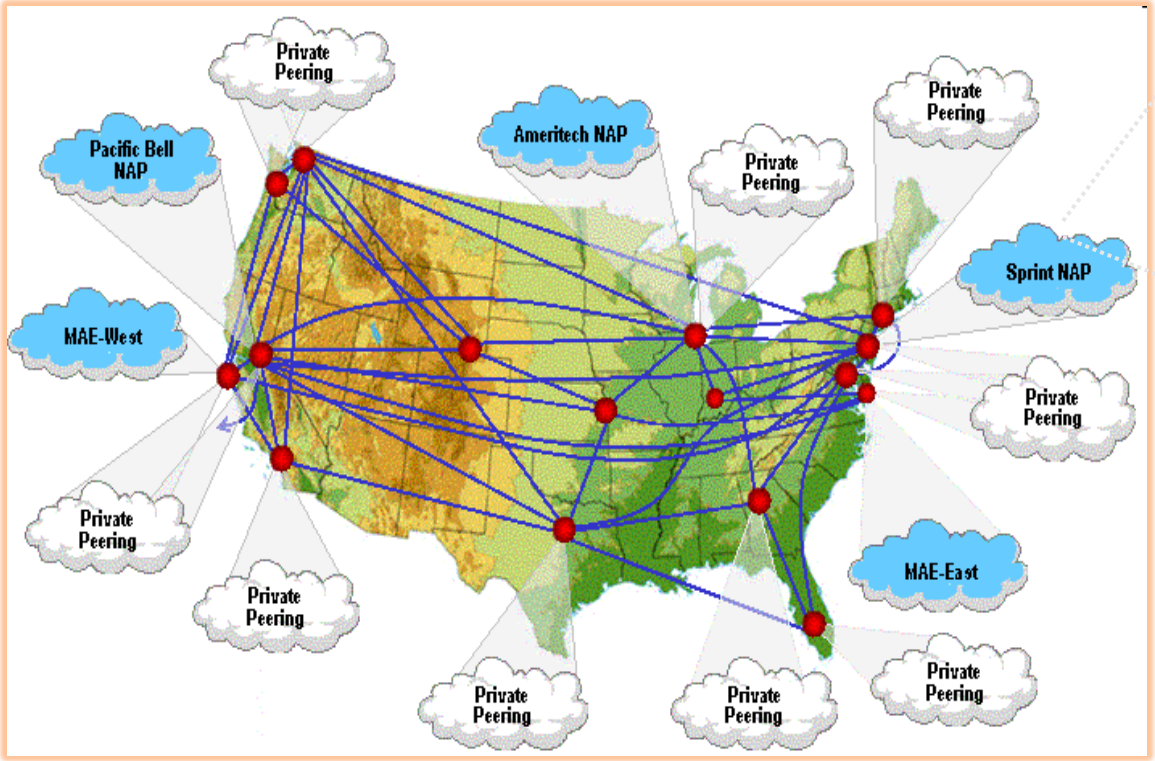
# THE **NETWORK CORE** CONSISTS OF ROUTERS AND COMMUNICATION LINKS THAT CONNECT ALL HOSTS.



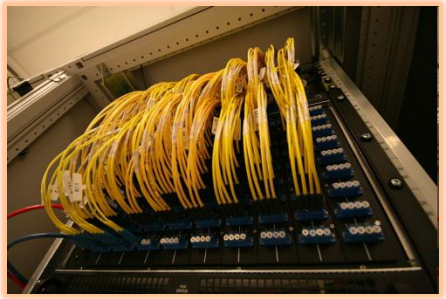

**ROUTERS** are specialized computers that manage network traffic.

Internet communication links are organized in a loose hierarchy similar to transport networks.

# THE **NET** HAS DECENTRALIZED STRUCTURE WITH DIFFERENT PARTS OWNED/MANAGED AUTONOMOUSLY.

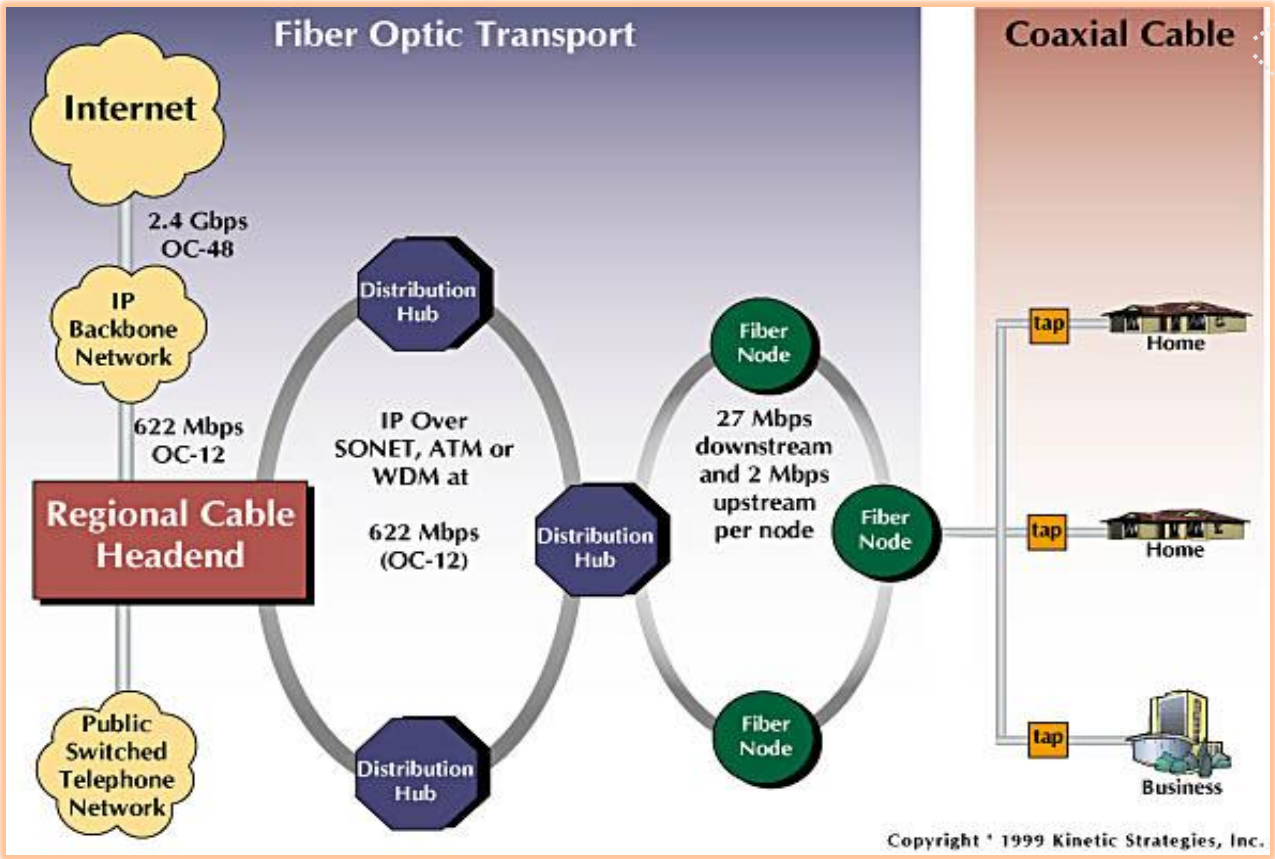



LINX (London)



Backbone Network Service Providers (NSPs) own long-haul networks and exchange data at ***INTERNET EXCHANGE POINTS*** (IXPs), *aka* ***NETWORK ACCESS POINTS*** (*NAP*) in the US.

# Regional/local ISPs deliver local traffic & aggregate traffic for long-haul delivery.



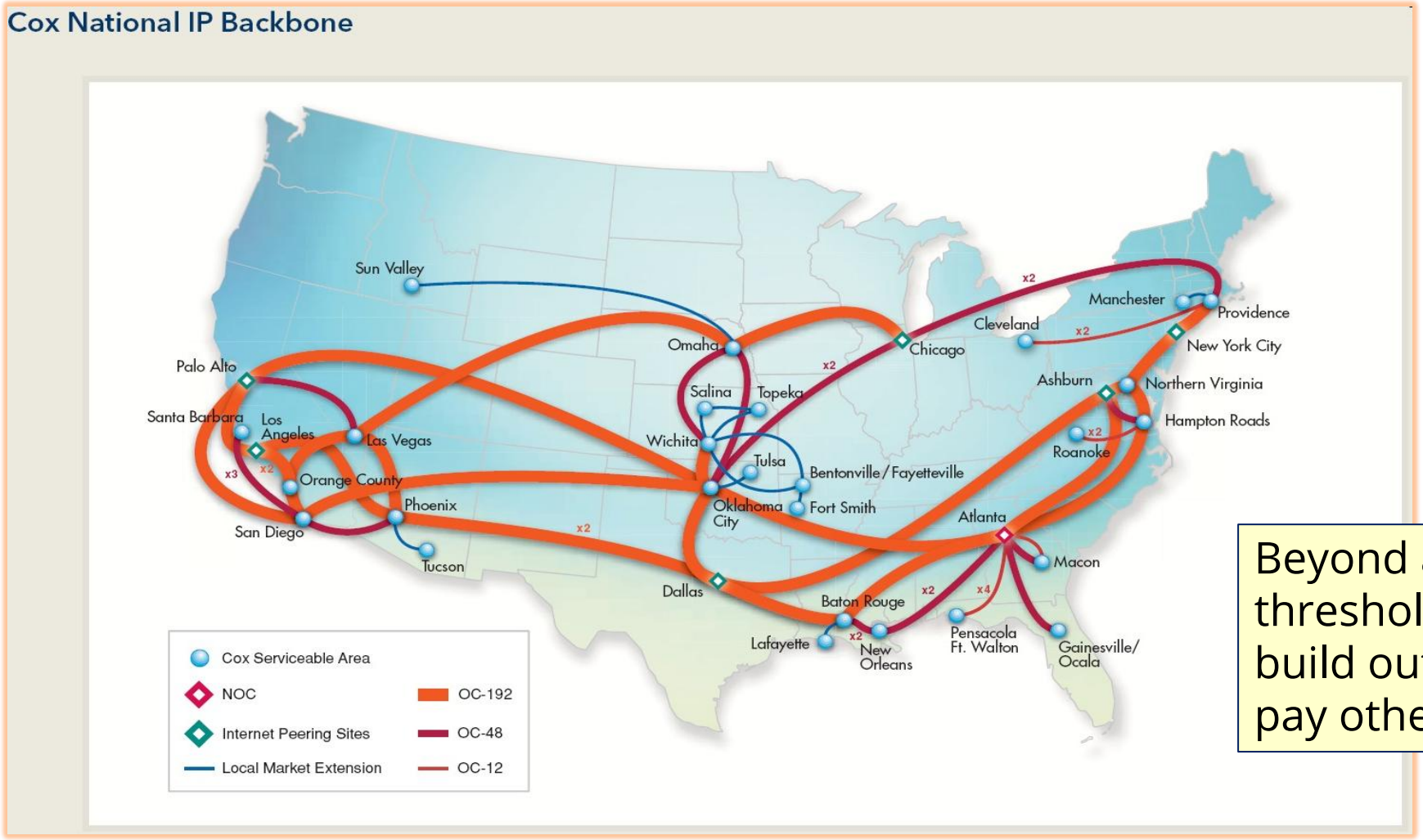
New installations use fiber-optic.

Long-haul fiber links are expensive; traffic aggregation ensures proper utilization.

# Observing Network Routes: traceroute

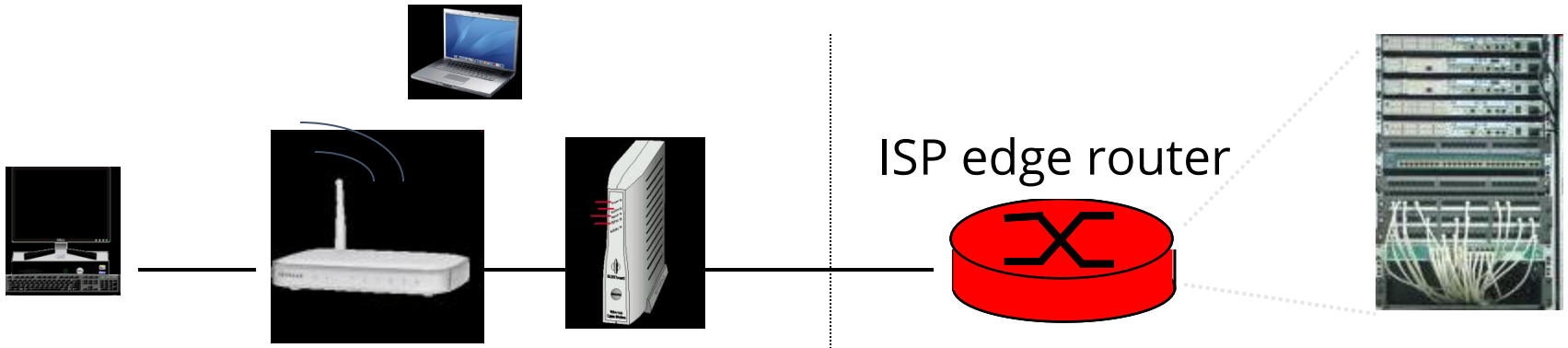
```
Tracing route to internet-service.ucc.uno.edu [137.30.1.92]
over a maximum of 30 hops:

  1     9 ms     9 ms     8 ms  10.128.32.1
  2    11 ms    10 ms    10 ms  68.11.12.25
  3    11 ms     8 ms    12 ms  mctydsrc01-gew0304.rd.no.cox.net [68.11.14.9]
  4    13 ms     9 ms    13 ms  mctybbrc01-pos0101.rd.no.cox.net [68.1.0.64]
  5    10 ms    10 ms     9 ms  mctybbrc02-pos0100.rd.no.cox.net [68.1.0.63]
  6    21 ms    19 ms    20 ms  lkhnbbrc02-pos0102.rd.at.cox.net [68.1.0.10]
  7    22 ms    21 ms    20 ms  lkhnbbrc01-pos0100.rd.at.cox.net [68.1.0.2]
  8    29 ms    21 ms    21 ms  so-1-2-0-0.gar2.Atlanta1.Level3.net [65.59.222.5]
  9    21 ms    21 ms    21 ms  so-0-3-0.bbr2.Atlanta1.Level3.net [209.247.11.225]
 10    35 ms    36 ms    34 ms  so-0-0-0.bbr1.Washington1.Level3.net [64.159.1.2]
 11    35 ms    37 ms    34 ms  so-6-0-0.edge1.Washington1.Level3.net [209.244.11.10]
 12    36 ms    35 ms    36 ms  qwest-level3-oc48.Washington1.Level3.net [209.244.219.182]
 13    35 ms    35 ms    36 ms  205.171.251.33
 14    35 ms    35 ms    37 ms  dca-core-02.inet.qwest.net [205.171.8.221]
 15    36 ms    37 ms    36 ms  atl-core-02.inet.qwest.net [205.171.8.153]
 16    37 ms    46 ms    37 ms  atl-edge-05.inet.qwest.net [205.171.21.54]
 17    51 ms    52 ms    51 ms  65.112.33.250
 18    56 ms    56 ms    60 ms  nor2-isp2.atm-vcc.La.Net [162.75.7.78]
 19     *        *        *     Request timed out.
```

# Regional ISPs increasingly use long-haul links for internal traffic delivery.

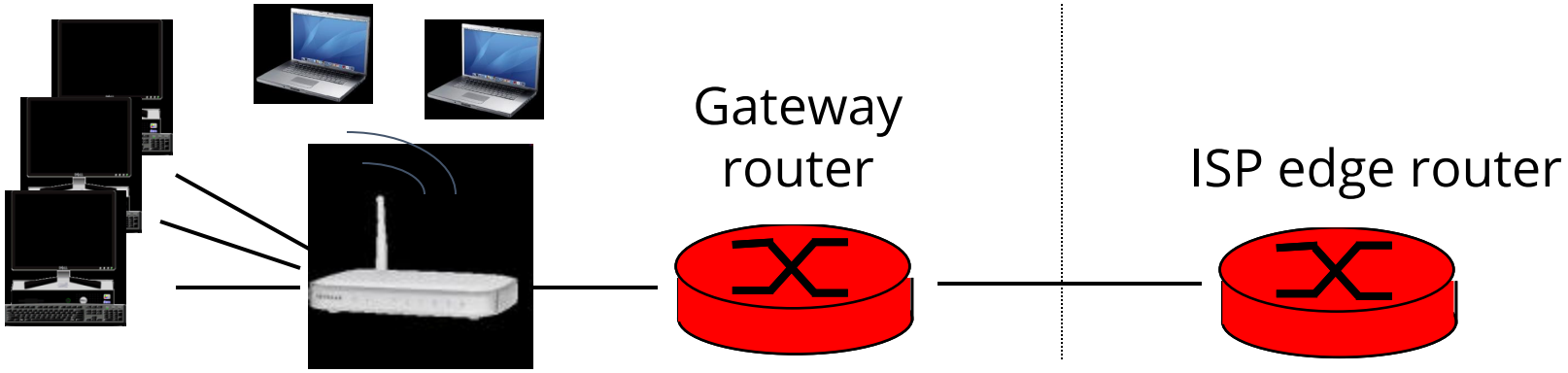



Beyond a certain (traffic) threshold, it is cheaper to build out the network then to pay other ISPs for delivery.

# END USERS CONNECT VIA LOCAL/REGIONAL ISP.

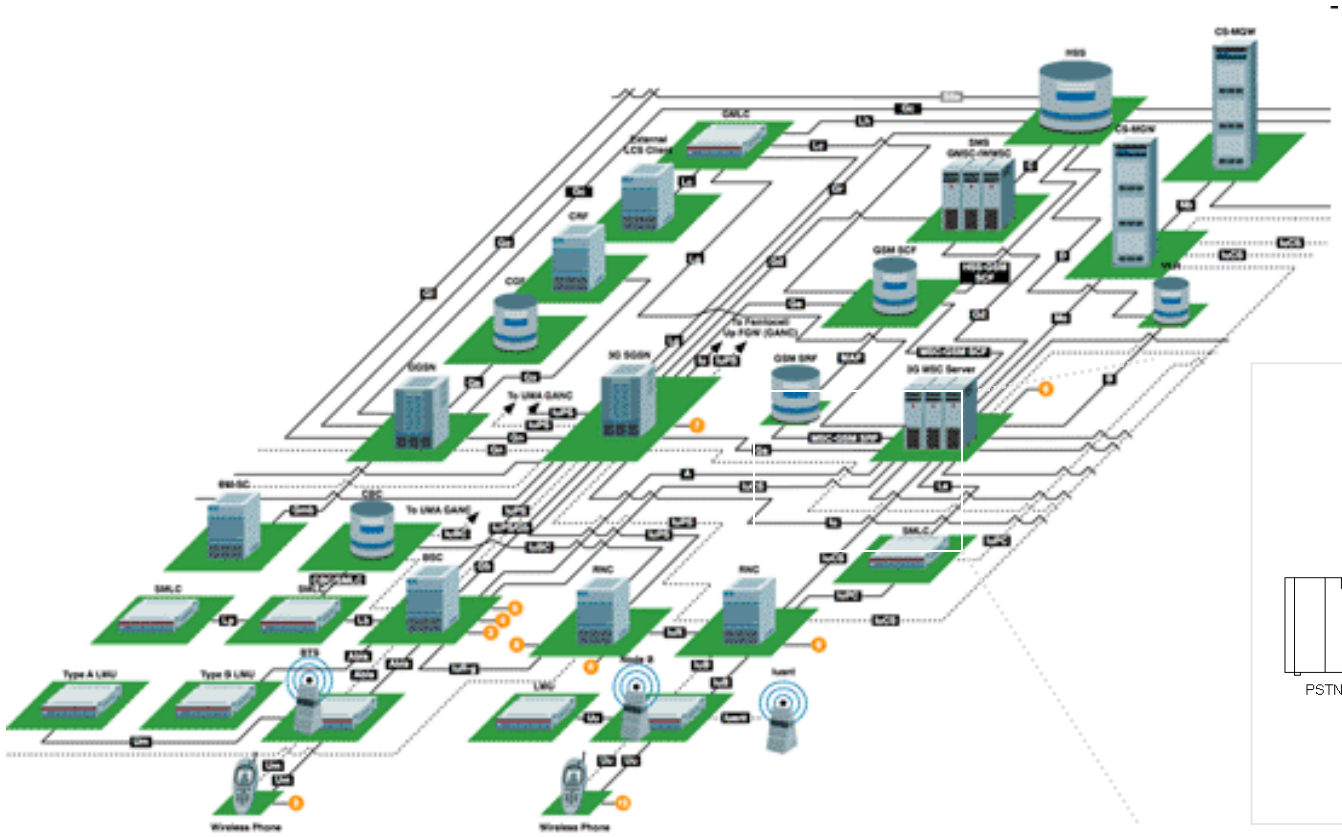


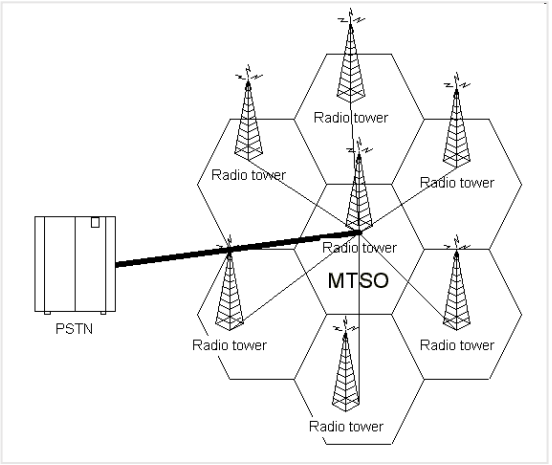

Residential access: cable/DSL modem



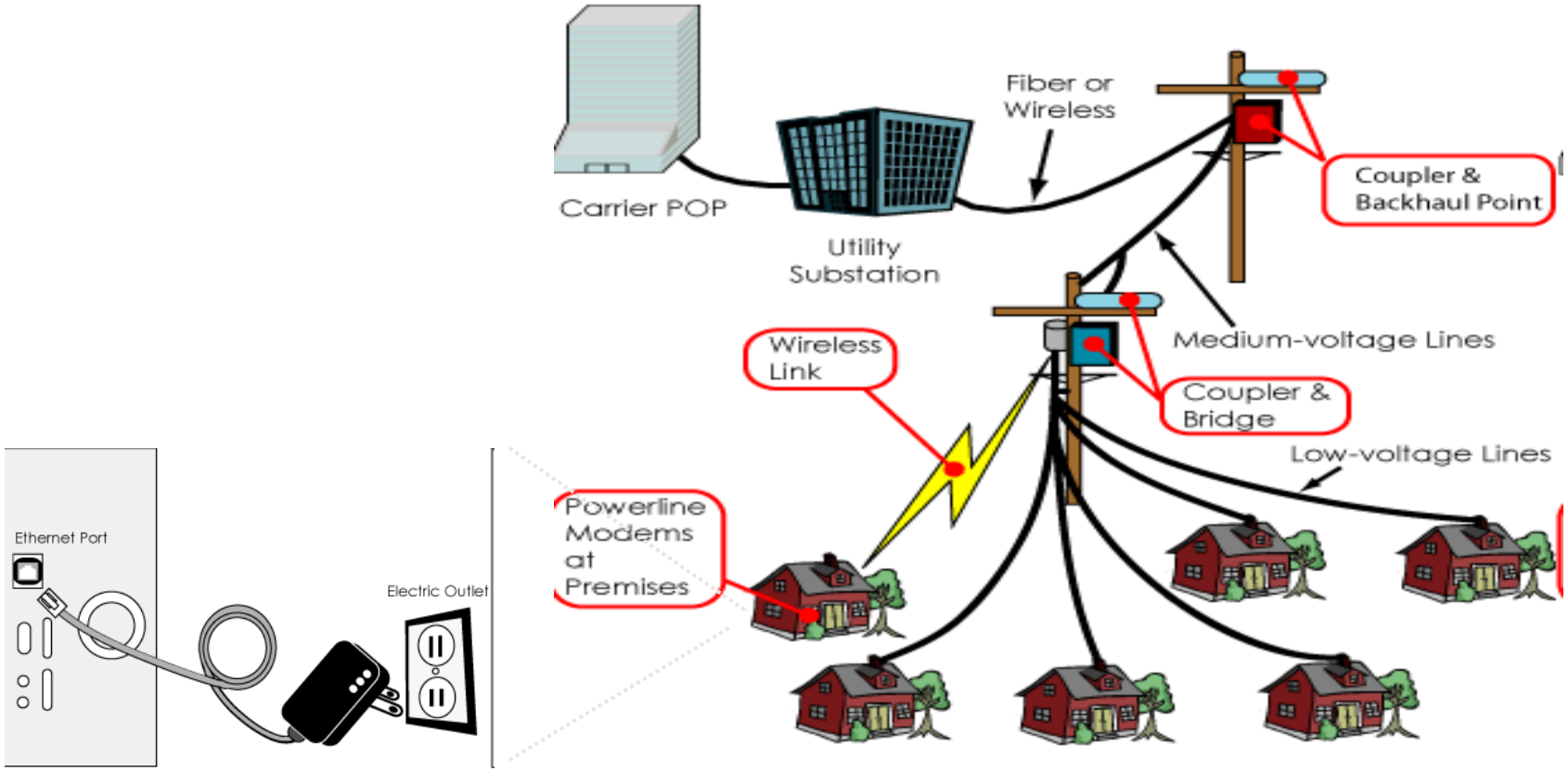

Institutional access: direct connection

# Cell network providers offer wide-area connectivity at lower transmission rates.





# Broadband over Power Lines (BPL) piggybacks on the electricity grid.



# Network protocols & apps

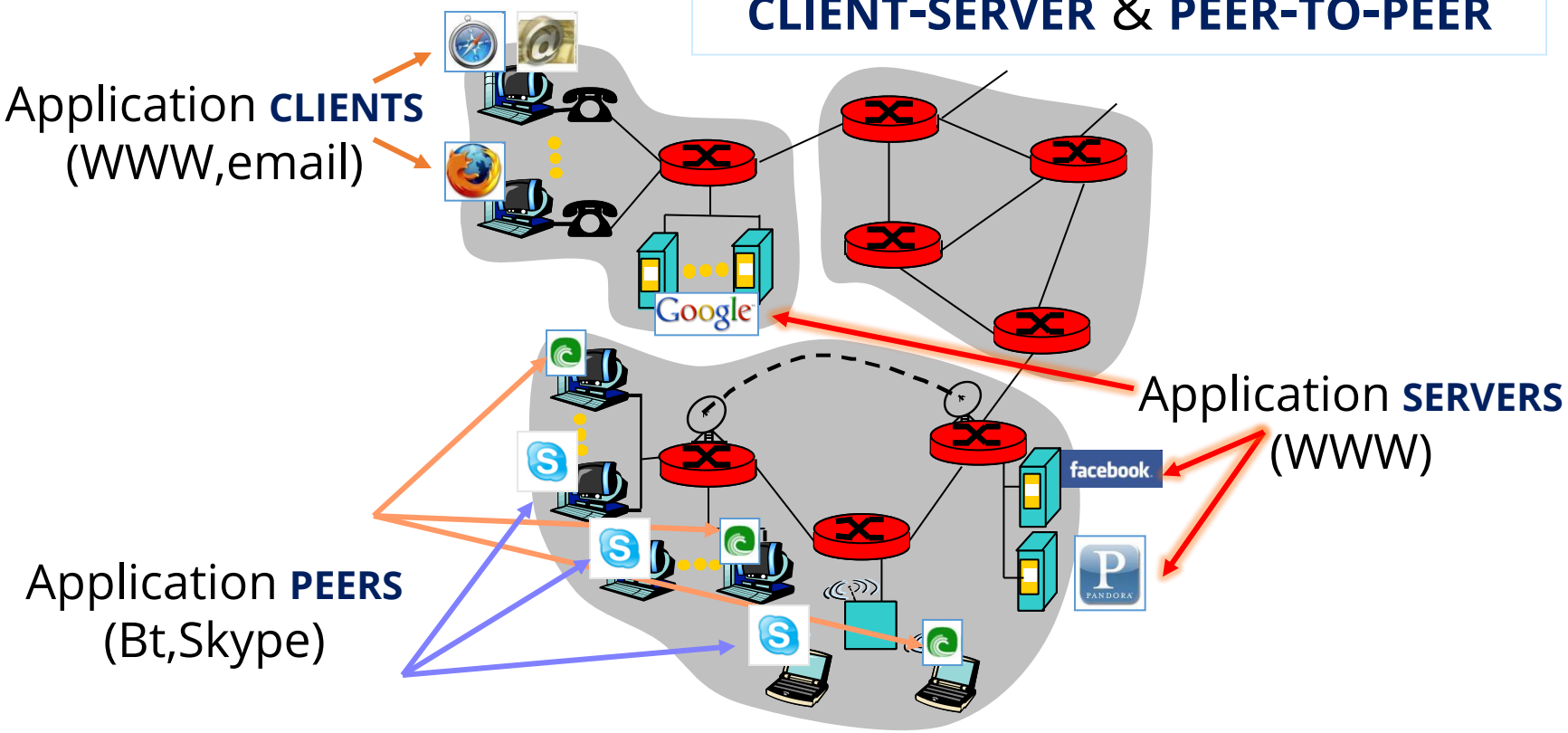
# Q: Why do we connect to the Internet?

## A: To use different network applications

All network applications rely on the same set of standard network communication services.

# NETWORK APPLICATIONS CONSIST OF TWO, OR MORE, PROCESSES COMMUNICATING OVER THE NETWORK.

Two application architectures:
**CLIENT-SERVER** & **PEER-TO-PEER**




**Each application defines its own protocol for exchanging messages; multiple implementations are possible.**

# TRY THIS AT A SHELL PROMPT:

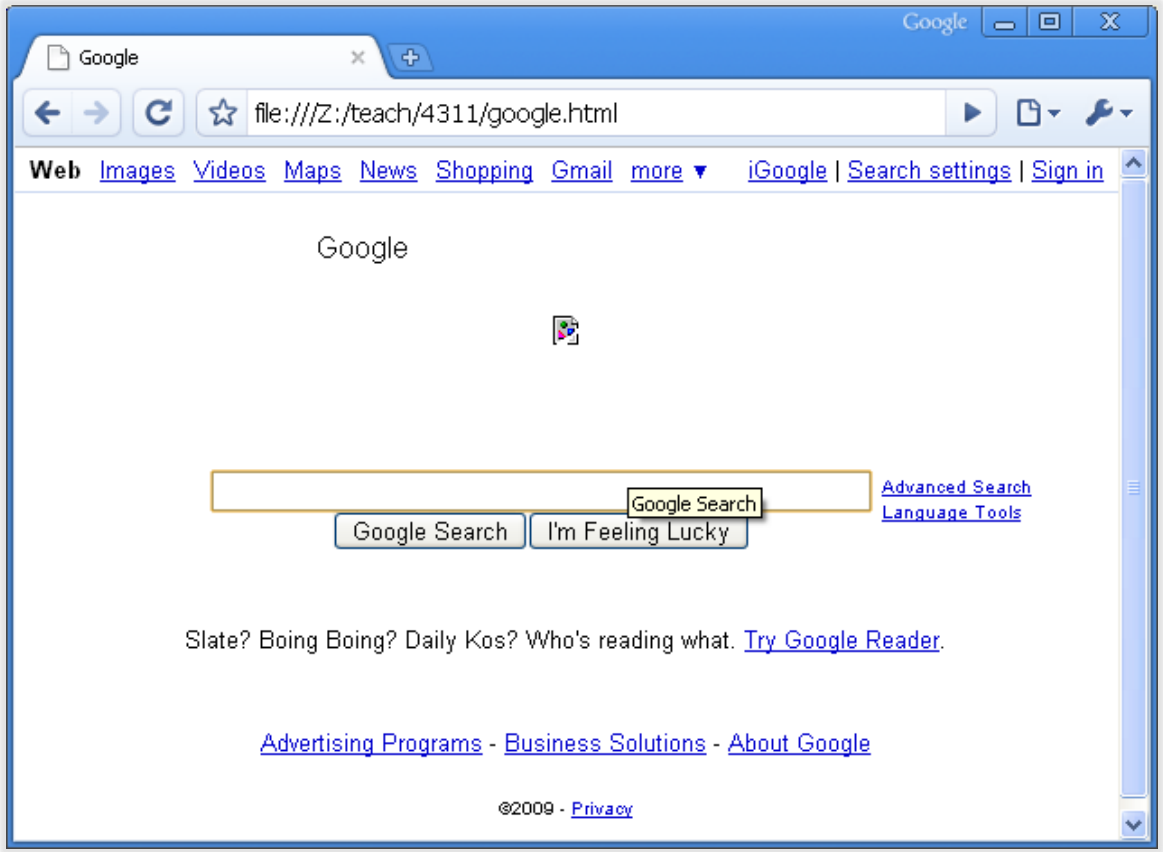
```
>telnet google.com 80 > google.html↵
GET / HTTP/1.0↵
↵
Connection closed by foreign host.
>_
```

# GOOGLE.HTML SHOULD LOOK LIKE THIS:

```
Trying 74.125.127.100...
Connected to google.com.
Escape character is '^]'.
HTTP/1.0 200 OK
Date: Wed, 02 Sep 2009 21:39:00 GMT
Expires: -1
Cache-Control: private, max-age=0
Content-Type: text/html; charset=ISO-8859-1
Set-Cookie: PREF=ID=28e4088ee4e9eb4b:TM=1251927540:LM=12
MT; path=/; domain=.google.com
Set-Cookie: NID=26=gkijYBJQ-o_lcBItp1xNqy4enRCW9k4aNIA7d
H12-IRASV81x3YLylObeCsjkDli; expires=Thu, 04-Mar-2010 21
Server: gws

<!doctype html><html><head><meta http-equiv="content-typ
<script>window.google={kEI:"9OWeSt2NL4LStgPp6cjmDQ",kEXP
WeSt2NL4LStgPp6cjmDQ"},kHL:"en"};
```

**Remove these lines, save, and open in browser**

# Browser view




**Congratulations—you just 'spoke' HTTP!**

# The World Wide Web (WWW)

Communication protocol

Network application

Document naming

Document format

**URL** + **HTTP** + **HTML** = **WWW**

HTTP 1.0: RFC 1945
HTTP 1.1: RFC 2068

PC web **client**
*e.g., Firefox*

HTTP request

HTTP response

HTTP request

HTTP response

Mac web
**client**
*Safari*

Linux web **server**
e.g. *Apache*

# Uniform Resource Locator (URL)

Protocol:
`http,ftp,…`

Server port
(80 for WWW)

`http://www.cs.uno.edu:80/index.html`

Server domain name/
IP address

Local object name
(opaque))

**Interpretation:**

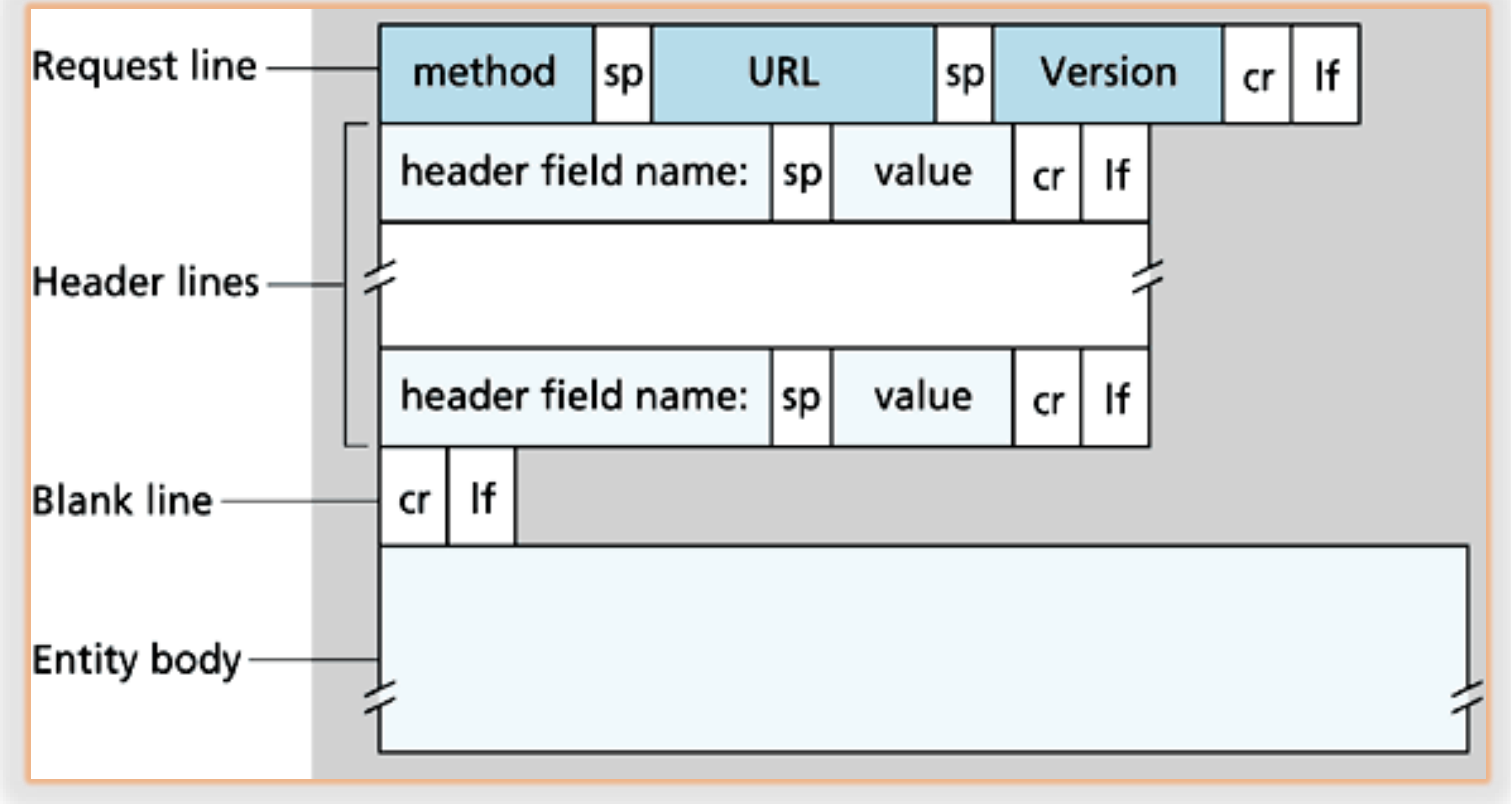
- Open TCP socket to `www.cs.uno.edu:80`
- Request `/index.html` using the `http` protocol

# HTTP REQUEST FORMAT

```
GET, POST, HEAD (1.0)
PUT, DELETE     (1.1)
```

```
/index.html
```

```
HTTP/1.0 or HTTP/1.1
```

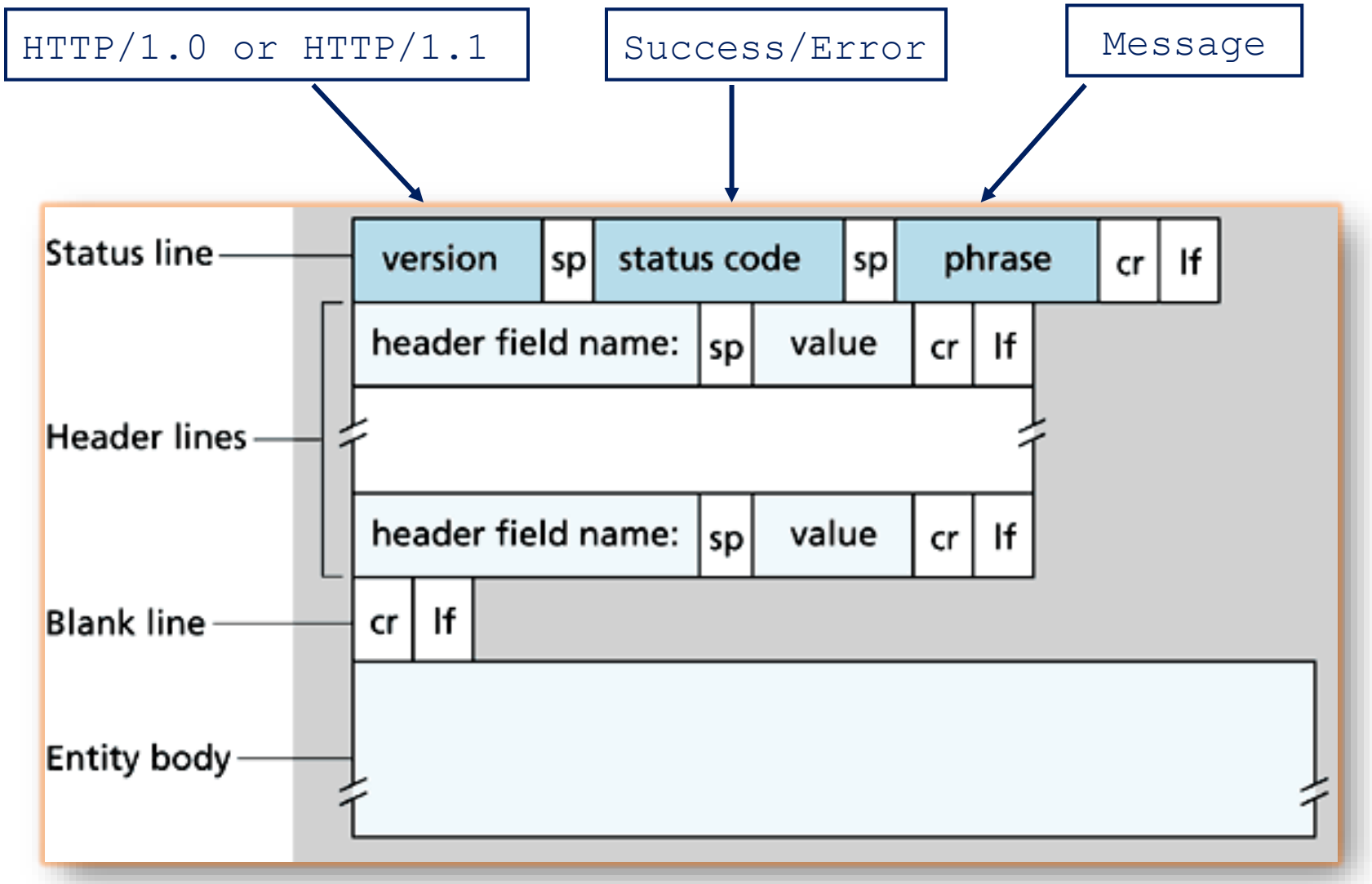


# HTTP REQUEST EXAMPLES

```
GET /~vassil HTTP/1.0
Connection: Keep-Alive
User-Agent: Mozilla/4.74 [en] (WinNT; U)
Host: dilbert.cs.uno.edu:8080
Accept: image/gif, image/x-xbitmap, image/jpeg,
        image/pjpeg, image/png, */*
Accept-Encoding: gzip
Accept-Language: en
Accept-Charset: iso-8859-1,*,utf-8
Cookie: SITESERVER=ID=8a064b7855a043146e45991174a3d970
```

```
GET /~vassil HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg,
        image/pjpeg, application/msword,
        application/vnd.ms-excel,
        application/vnd.ms-powerpoint, */*
Accept-Language: en-us
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 5.5; Windows NT 4.0)
Host: dilbert.cs.uno.edu:8080
Connection: Keep-Alive
```

# HTTP RESPONSE FORMAT



# HTTP RESPONSE EXAMPLE

status line (protocol status code status phrase)

header lines

blank line

data, e.g., requested HTML file

```
HTTP/1.1 200 OK
Connection close
Date: Thu, 06 Aug 1998 12:00:15 GMT
Server: Apache/1.3.0 (Unix)
Last-Modified: Mon, 22 Jun 1998 …...
Content-Length: 6821
Content-Type: text/html

<!doctype html><html><head><meta …
```

# HTTP RESPONSE TYPES (RFC 2616)

**2xx Success**

request succeeded, requested object later in this message

**3xx Redirection**

requested object moved, new location specified later in this message

**4xx Bad client request**

request message not understood by server

**5xx Server error**

an error occurred while processing the request

# Common response status codes

**200 OK**

- request succeeded, requested object later in this message

**301 Moved Permanently**

- requested object moved, new location specified later in this message (Location:)

**400 Bad Request**

- request message not understood by server

**404 Not Found**

- requested document not found on this server

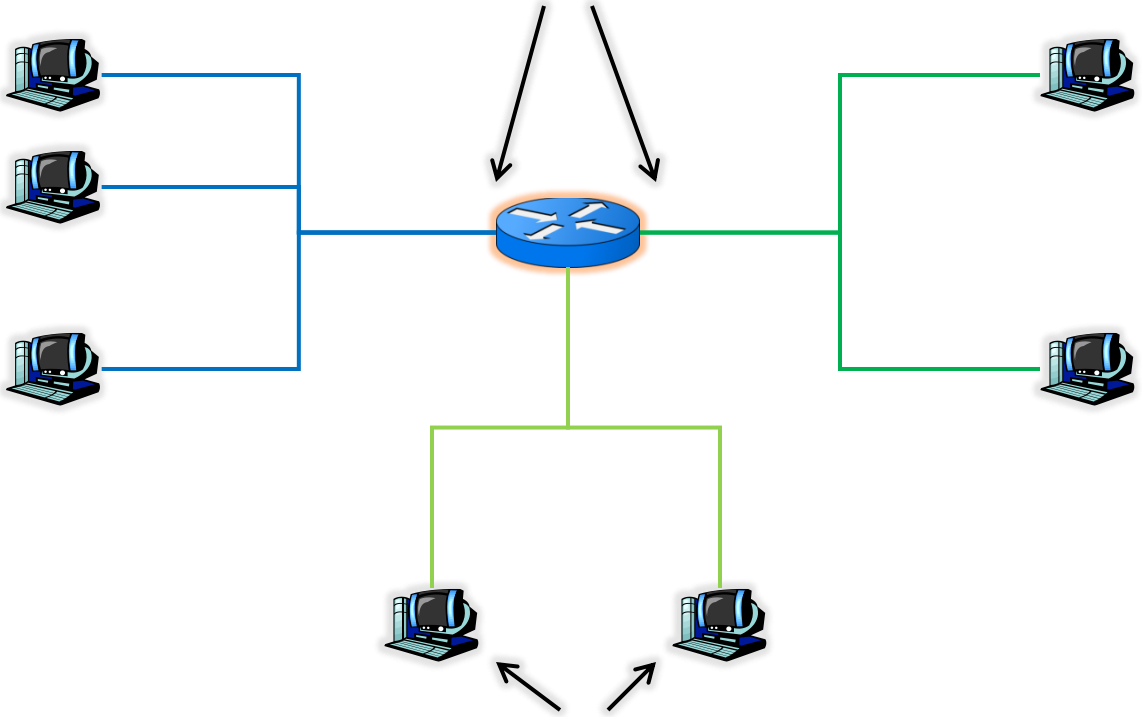
**505 HTTP Version Not Supported**

# IP ADDRESSES

**NETWORK INTERFACE:**

A CONNECTION BETWEEN HOST OR ROUTER AND A PHYSICAL LINK.

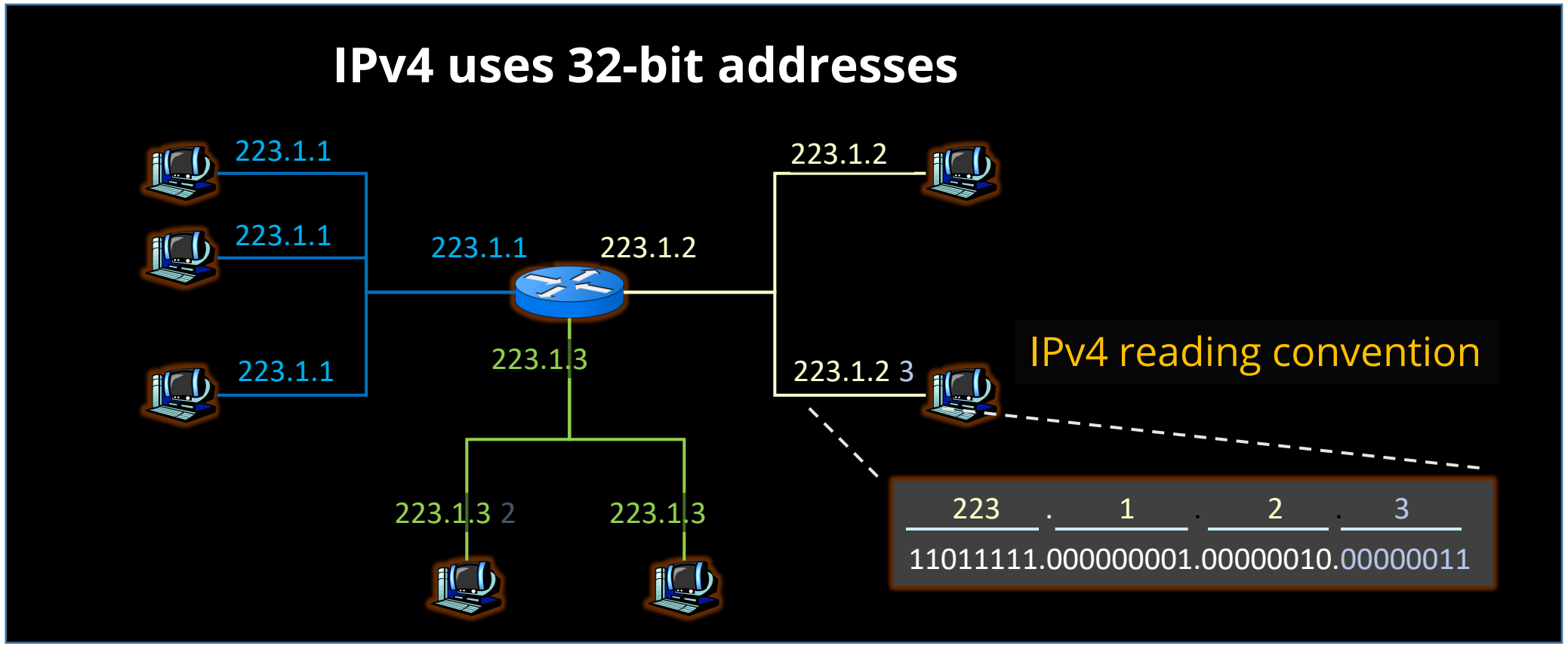
Routers have multiple interfaces



Hosts typically have one interface

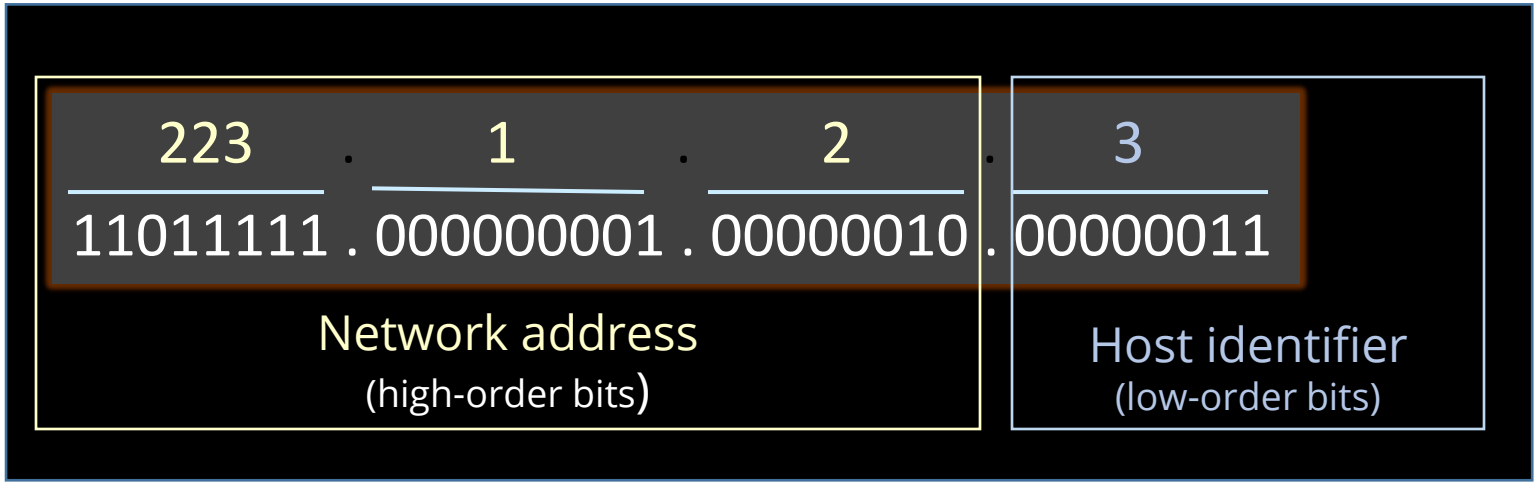
# IP ADDRESS:

A UNIQUE NUMBER THAT IDENTIFIES A NETWORK INTERFACE & CONTAINS ADDRESSING INFORMATION ON HOW TO FIND THE HOST.



**LOCAL AREA NETWORK**: THE SET OF DEVICES THAT CAN COMMUNICATE WITH EACH OTHER WITHOUT AN INTERVENING ROUTER (ATTACHED TO SAME PHYSICAL NETWORK).

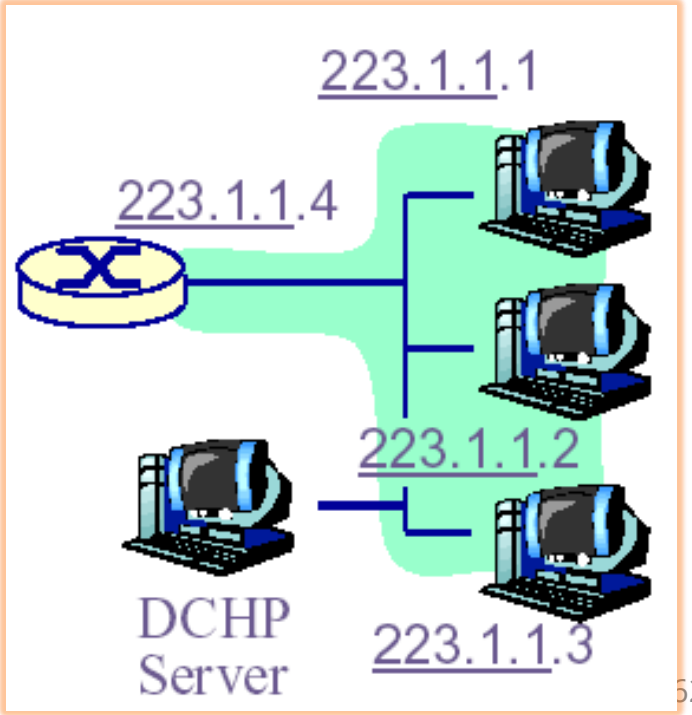
**IP network:** The set of device interfaces with IP addresses having a common network part



# IP Address Assignment

- Network address is assigned by the ISP
- Host address
  - Static assignment
    - Configuration parameter (manually) set during system installation
  - Dynamic assignment at boot/wake-up time

    DHCP: Dynamic Host Configuration Protocol:
    - Host broadcasts a “DHCP discover” message
    - DHCP server responds with a “DHCP offer” message
    - Host requests IP address: “DHCP request” message
    - DHCP server sends address: “DHCP ack” message



# Private IPv4 Address Spaces (RFC 1918)

- 10.0.0.0–10.255.255.255
  - Single class A network; 10.0.0.0/8 (255.0.0.0)
  - 24-bit host id (16,777,216 addresses)
- 172.16.0.0–172.31.255.255
  - 16 contiguous class B; 172.16.0.0/12 (255.240.0.0)
  - 20-bit host id (1,048,576 addresses)
- 192.168.0.0–192.168.255.255
  - 256 contiguous class C; 192.168.0.0/16 (255.255.0.0)
  - 16-bit host id (65,536 addresses)

# IPv6

- Initial motivation:
  - 32-bit address space currently completely allocated
- Additional motivation:
  - Header format helps speed processing/forwarding
  - Header changes to facilitate QoS
  - New “anycast” address: route to “best” of several replicated servers
- IPv6 datagram format:
  - Fixed-length 40 byte header
  - No fragmentation allowed

# Text Representation of Addresses

- Preferred form

  `1080:0:FF:0:8:800:200C:417A`

- Compressed

  `FF01:0:0:0:0:0:0:43` → `FF01::43`

- IPv4-compatible

  `0:0:0:0:0:0:13.1.68.3`, or `::13.1.68.3`

- URL

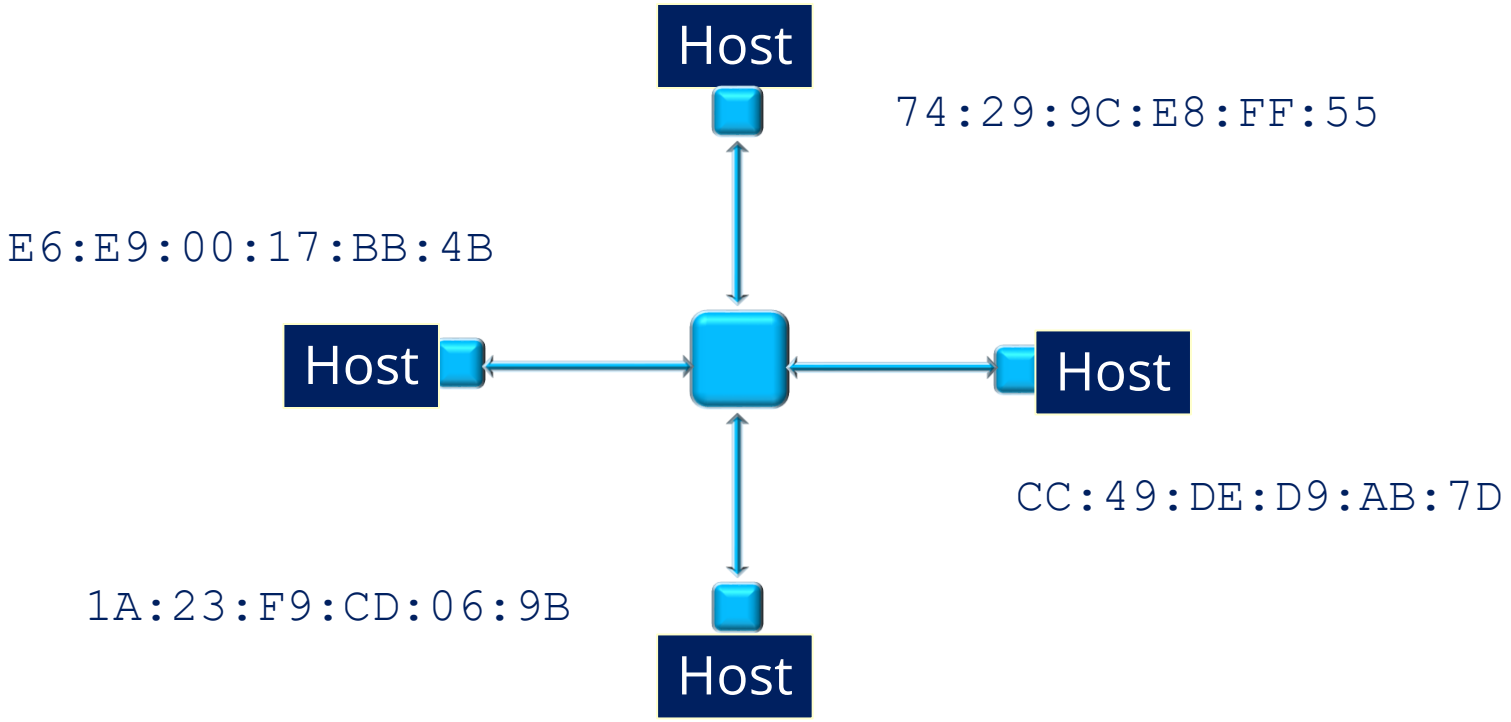
  `http://[FF01::43]/index.html`

# MAC/PHYSICAL ADDRESSES

# Ethernet/physical MAC Address

- **MAC**: **M**edium **A**ccess **C**ontrol
- IP address:
  - **logical 32-bit network-layer address**
  - **assigned at start up**
  - used to get datagram to destination network
- Ethernet/802.11 or MAC or physical) address:
  - **permanent 48-bit MAC address:** (burned in the adapter's ROM)
  - used to get datagram from one interface to another physically-connected interface (same network)

# Ethernet Addresses




Each adapter has a globally unique address.

# Ethernet MAC Address

- Ethernet MAC address allocation administered by IEEE
- Manufacturer buys portion of MAC address space (to assure uniqueness)
- MAC address is unstructured ➔ portable
  - i.e., can move adapter from one *Ethernet* to another
- IP hierarchical address NOT portable
  - depends on network to which one attaches

# Concept index

- Internet structure
- Network protocols
  - Example: WWW
- IP addresses
- MAC addresses